



YEDITEPE UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

FINITE ELEMENT ANALYSIS OF THE DEEP DRAWING PROCESS USING AN
AXISYMMETRIC MODEL

A Thesis Submitted

by

Serdar Alkın Bilge

Mertcan Aras

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science
in
Mechanical Engineering

Supervisor

Assoc. Prof. Dr. Ali Fethi Okyar

Istanbul

FINITE ELEMENT ANALYSIS OF THE DEEP DRAWING PROCESS USING AN
AXISYMMETRIC MODEL

by
Serdar Alkın Bilge
Mertcan Aras

Approved by:

Assoc. Prof.. Dr. Ali Fethi Okyar
(Yeditepe University)
(Thesis Supervisor)

.....

DATE OF APPROVAL:/...../20...

DECLARATION OF ORIGINALITY

I hereby declare that this thesis is my own work and that all information in this thesis has been obtained and presented following academic rules and ethical conduct. I have fully cited and referenced all material and results as required by these rules and conduct, and this thesis study does not contain any plagiarism. The necessary permissions have been obtained if any material used in the thesis requires copyright. No material from this thesis has been used to award another degree.

To the best of my knowledge and belief, it contains no material previously published or written by another person nor material accepted for the award of any other degree except where due acknowledgment has been made in the text.

I accept all kinds of legal liability that may arise in cases contrary to these situations.

Serdar Alkın Bilge

Mertcan Aras

Signature

ABSTRACT

This thesis investigates the mechanical influence of manufacturing-induced cold-working on the structural integrity of deep-drawn components using a high-fidelity 2D axisymmetric finite element (FE) framework. In modern manufacturing, neglecting the complete deformation history such as non-uniform thinning, residual stresses, and plastic strain accumulation leads to significant inaccuracies in the subsequent mechanical evaluation of deep-drawn vessels.

The numerical methodology utilizes an integrated open-source ecosystem comprising Gmsh for parametric meshing, PrePoMax for pre-processing, and CalculiX for nonlinear elastoplastic solving. A flat 1060 Aluminum alloy blank was modeled using second-order quadratic axisymmetric continuum elements (CAX8R) to overcome volumetric locking and accurately capture steep stress gradients at critical tool radii. The material behavior was characterized by a manually calibrated isotropic hardening curve, ranging from an initial yield strength of 30 MPa to a saturation point of 70 MPa.

To enhance forming stability and minimize the risk of wrinkling or tearing, a sophisticated Four-Step Incremental Solution Strategy was implemented. This workflow includes a specialized 3 mm blank holder retraction phase (based on the Manabe friction-reduction criterion) to allow for stress relaxation and more homogeneous material flow before the final drawing stage.

The results demonstrate that the peak Von Mises stress and Equivalent Plastic Strain (PEEQ) are highly localized at the punch radius, representing the most heavily cold-worked zones. By analyzing the full unloading phase, the study quantifies the spring-back effect and maps the permanent "as-manufactured" mechanical footprint. This research concludes that utilizing this realistic deformed state, rather than idealized CAD geometry, is paramount for the accurate prediction of a vessel's ultimate performance under service loads, such as internal pressure.

Keywords: Deep Drawing, Finite Element Analysis, CalculiX, 1060 Aluminum, Cold-Working, Residual Stress, Axisymmetric Modeling.

ÖZET

Bu çalışma, derin çekme yöntemiyle üretilen bileşenlerin yapısal bütünlüğü üzerinde, üretim kaynaklı soğuk işlemin (cold-working) mekanik etkilerini yüksek sadakatli 2D eksenel simetrik sonlu elemanlar (FE) çerçevesi kullanarak incelemektedir. Modern imalatta; homojen olmayan inceltme, kalıntı gerilmeler ve plastik gerinim birikimi gibi üretim geçmiş verilerinin ihmal edilmesi, derin çekme kapların servis yükleri altındaki mekanik değerlendirmelerinde önemli hatalara yol açmaktadır.

Sayısal metodoloji; parametrik ağ yapısı için Gmsh, ön işlemci olarak PrePoMax ve lineer olmayan elastoplastik çözümleyici olarak CalculiX yazılımlarından oluşan entegre bir açık kaynak ekosistemini kullanmaktadır. Düz bir 1060 Alüminyum alaşımlı sac taslak, hacimsel kilitlenme (volumetric locking) sorununu aşmak ve kritik takım yarıçaplarındaki dik gerilme gradyanlarını doğru yakalamak amacıyla ikinci mertebe biquadratic eksenel simetrik kontinyum elemanlar (CAX8R) ile modellenmiştir. Malzeme davranışı, 30 MPa başlangıç akma dayanımından 70 MPa doyma noktasına kadar uzanan, manuel olarak kalibre edilmiş bir izotropik pekleşme eğrisi ile tanımlanmıştır.

Şekillendirme stabilitesini artırmak, yırtılma ve kırılma riskini minimize etmek için dört aşamalı, gelişmiş bir çözüm stratejisi uygulanmıştır. Bu iş akışı, Manabe sürtünme azaltma kriterine dayanarak, gerilmelerin yeniden dağıtılmasına ve malzemenin daha homojen bir şekilde akmasına olanak tanıyan 3 mm'lik baskı plakası (blank holder) geri çekme aşamasını içermektedir.

Analiz sonuçları, en yüksek Von Mises gerilmesi ve Eşdeğer Plastik Gerinim (PEEQ) değerlerinin zımba (punch) yarıçapında yoğunlaştığını ve bu bölgenin en yoğun soğuk işleme maruz kalan alan olduğunu göstermektedir. Tam boşaltma (unloading) aşamasının incelenmesiyle, geri esneme (spring-back) etkisi nicelleştirilmiş ve üretilen parçanın kalıcı mekanik ayak izi haritalandırılmıştır. Bu araştırma, derin çekme kapların patlama basıncı veya servis yükleri altındaki performansının doğru öngörülebilmesi için idealize edilmiş CAD geometrileri yerine, bu gerçekçi deforme olmuş durumun başlangıç koşulu olarak kabul edilmesinin hayati önem taşıdığını ortaya koymaktadır.

Anahtar Kelimeler: Derin Çekme, Sonlu Elemanlar Analizi, CalculiX, 1060 Alüminyum, Soğuk İşlem, Kalıntı Gerilme, Eksenel Simetrik Modelleme

ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to our supervisor, Assoc. Prof. Dr. Ali Fethi Okyar, for his invaluable guidance, technical expertise, and continuous encouragement throughout this project. His profound insights into nonlinear mechanics and finite element formulations have been fundamental in shaping the methodology and ensuring the success of this research.

We are truly grateful to our families, whose unwavering support, patience, and motivation provided us with the strength to navigate the complexities of this graduation thesis.

Lastly, we would like to express our heartfelt appreciation for each other as teammates. Collaborating on the manual calibration of material models, implementing advanced boundary conditions, and conducting multi-step FEA simulations has been a highly productive and rewarding experience. This project is a testament to our mutual dedication and collective effort.

To everyone who contributed directly or indirectly to this study, we extend our sincere thanks.

TABLE OF CONTENTS

DECLARATION OF ORIGINALITY	iii
ABSTRACT.....	iv
ÖZET	v
ACKNOWLEDGEMENTS.....	vi
TABLE OF CONTENTS.....	vii
LIST OF FIGURES	ix
LIST OF TABLES.....	x
LIST OF ABBREVIATIONS.....	xi
1. INTRODUCTION	1
2. LITERATURE REVIEW	4
2.1. Analytical and Experimental Investigations.....	4
2.2. FINITE ELEMENT MODELLING OF DEEP DRAWING.....	5
2.3. EFFECT OF PROCESS PARAMETERS AND MATERIAL BEHAVIOUR..	6
2.4. SPRING-BACK AND POST-FORMING GEOMETRY	6
2.5. VON MISES YIELD CRITERION AND STRESS THEORY	7
3. METHODOLOGY	8
3.1. NUMERICAL MODELLING STRATEGY	8
3.2. FINITE ELEMENT FORMULATION AND ELEMENT SELECTION.....	8
3.2.1. CAX: Axisymmetric Continuum Formulation	9
3.2.2. 8: Higher-Order (Quadratic) Interpolation	9
3.2.3. R: Reduced Integration Technique	10
3.2.4. Synthesis of the High-Fidelity Selection	10
3.3. SOLUTION PROCEDURE AND STEP DEFINITIONS.....	11
3.3.1. Step 1: Clamping and Initial Forming (0–5 mm)	12
3.3.2. Step 2: Boundary Condition Shift (Holder Retraction)	12
3.3.3. Step 3: Final Forming in Unconstrained State (5–10 mm).....	12

3.3.4. Step 4: Unloading and Spring-back Identification	13
3.4. NUMERICAL FORMULATION AND COUPLING	13
4. ANALYSIS AND RESULTS.....	14
4.1. Verification of Manual Element Transformation (CAX8R)	14
4.2. EQUIVALENT TOTAL STRAIN ANALYSIS	16
4.3. STRESS ANALYSIS (VON MISES STRESS).....	19
4.4. Equivalent Plastic Strain Analysis (PEEQ).....	22
4.5. Elastic Strain Energy Analysis	24
4.6. Displacement Analysis	27
4.7. CONTACT PRESSURE ANALYSIS.....	29
4.8. CONTACT SLIP ANALYSIS	32
5. DISCUSSION.....	35
5.1. MATERIAL CHARACTERIZATION AND HARDENING FIDELITY	35
5.2. IMPLEMENTATION OF ADVANCED BOUNDARY CONDITIONS: THE MANABE CRITERION.....	35
5.2.1. Friction Dynamics and Forming Stability (Singh Analysis)	36
5.3. GEOMETRIC OPTIMIZATION AND FAILURE MITIGATION (CHEN APPROACH).....	36
5.4. NUMERICAL INTEGRITY AND MESH SENSITIVITY ANALYSIS	37
5.5. STRATEGIC IMPLICATIONS OF MANUFACTURING-INDUCED RESIDUAL STRESSES.....	37
6. CONCLUSION.....	38
7. REFERENCES	40
8. APPENDIX.....	42

LIST OF FIGURES

Figure 1. Deformation zones and force distribution in the deep drawing process.	2
Figure 2. Deep drawing process , (a) drawing a cup, (b) process variables.	3
Figure 3. 1060 Aluminum Stress Strain Curve.....	5
Figure 4. CAX8R.....	10
Figure 5. Initial 2D axisymmetric geometric model of the deep drawing setup (punch, die, holder, and blank) created in Gmsh, prior to mesh discretization.	11
Figure 6. Quadratic node distribution on the blank holder assembly.	14
Figure 7. Quadratic node distribution on the punch assembly.	15
Figure 8. Close-up view of the quadratic node distribution on the 1060 Aluminum blank.	15
Figure 9. Quadratic node distribution on the die assembly.	16
Figure 10. Total Strain Step 1	17
Figure 11. Total Strain Step 2	18
Figure 12. Total Strain Step 3	18
Figure 13. Total Strain Step 4	18
Figure 14. Total Strain Step 5	19
Figure 15. Von Mises Step 1	20
Figure 16. Von Mises Step 2	20
Figure 17. Von Mises Step 3	21
Figure 18. Von Mises Step 4	21
Figure 19. Von Mises Step 5	21
Figure 20. Step 1 PEEQ.....	22
Figure 21. Step 2 PEEQ.....	23
Figure 22. Step 3 PEEQ.....	23
Figure 23. Step 4 PEEQ.....	23
Figure 24. Step 5 PEEQ.....	24
Figure 25. Step 1 Elastic Strain Energy Analysis.....	25
Figure 26. Step 2 Elastic Strain Energy Analysis.....	25
Figure 27. Step 3 Elastic Strain Energy Analysis.....	26
Figure 28. Step 4 Elastic Strain Energy Analysis.....	26

Figure 29. Step 5 Elastic Strain Energy	26
Figure 30. Step 1 Displacement Analysis	28
Figure 31. Step 2 Displacement Analysis	28
Figure 32. Step 3 Displacement Analysis	28
Figure 33. Step 4 Displacement Analysis	29
Figure 34. Step 5 Displacement Analysis	29
Figure 35. Step 1 Contact Pressure Analysis	30
Figure 36. Step 2 Contact Pressure Analysis	30
Figure 37. Step 3 Contact Pressure Analysis	31
Figure 38. Step 4 Contact Pressure Analysis	31
Figure 39. Step 5 Contact Pressure Analysis	31
Figure 40. Step 1 Contact Slip Analysis	32
Figure 41. Step 2 Contact Slip Analysis	33
Figure 42. Step 3 Contact Slip Analysis	33
Figure 43. Step 4 Contact Slip Analysis	34
Figure 44. Step 5 Contact Slip Analysis	34

LIST OF TABLES

Table 1. Material Properties.....	9
Table 2. Geometrical Parameters.....	11
Table 3. Summary of the displacement-controlled operational steps.	13

LIST OF ABBREVIATIONS

Technical Abbreviations

FEA: Finite Element Analysis
 FEM: Finite Element Method
 CAD: Computer-Aided Design
 CAX8R: 8-node Biquadratic Axisymmetric Element with Reduced Integration
 PEEQ: Equivalent Plastic Strain
 TOSTRAIN: Equivalent Total Strain
 CPRESS: Contact Pressure
 CSLIP: Contact Slip
 ENER: Elastic Strain Energy
 LDR: Limiting Drawing Ratio
 BHF: Blank Holder Force
 DOF: Degrees of Freedom
 MPa: Megapascal (Unit of Pressure and Stress)

Mathematical Symbols

σ_{eq} : Von Mises Equivalent Stress
 σ : True Stress
 ε : Equivalent Plastic Strain
 K: Strength Coefficient.
 n: Strain-Hardening Exponent.
 τ : Frictional Shear Stress.
 μ : Friction Coefficient.
 P: Contact Pressure.
 F_{bh} : Blank Holder Force.
 T_0 : Initial Blank Thickness.
 W_{punch} : Punch Width.
 R_p : Punch Fillet Radius.
 R_d : Die Fillet Radius.
 $\sigma_r, \sigma_\theta, \sigma_z$: Radial, Hoop, and Axial Stress Components

1. INTRODUCTION

Deep drawing is one of the most essential sheet-metal forming processes in modern manufacturing, enabling the conversion of a flat metallic blank into a three-dimensional cup-like structure through the combined action of a punch, die, and blank holder [1, 8]. Owing to its high production efficiency, excellent material utilization, suitability for lightweight design, and capability to form complex geometries at relatively low cost, deep drawing is widely employed in automotive, aerospace, packaging, household appliances, and various structural applications [2, 3]. Despite its extensive industrial use, the deep drawing process remains inherently complex due to the nonlinear, path-dependent, and highly coupled deformation mechanisms involved during forming.

From a mechanical perspective, deep drawing constitutes a severe cold-working operation. During the forming process, the sheet metal is subjected to a combination of tensile, compressive, bending, and through-thickness stresses, which result in a highly non-uniform plastic deformation field [3, 7]. As a consequence, the material's initial mechanical state is fundamentally altered: plastic strain accumulates, residual stresses develop, anisotropy evolves, and most importantly, a non-uniform thickness distribution forms across the component wall [1, 5, 12]. These manufacturing-induced effects play a critical role in determining the final structural integrity of the part. Therefore, any post-forming mechanical assessment based solely on the initial blank thickness or an idealized CAD geometry is insufficient to capture the true behavior of deep-drawn components.

Historically, the design and optimization of deep drawing processes relied heavily on empirical rules and trial-and-error approaches. Early analytical tools, such as slip-line field theory and slab methods, provided limited predictive capability due to strong simplifications in geometry, loading conditions, and frictional behavior. With the rapid development of computational mechanics in the 1970s and 1980s, the Finite Element Method (FEM) emerged as a transformative tool for metal forming analysis [2, 6]. FEM enabled realistic modeling of key forming phenomena such as wrinkling, tearing, ironing, thickness evolution, and spring-back [1, 4, 13]. Continuous advancements in contact algorithms, friction formulations, and anisotropic yield criteria have since established FEM as an indispensable tool for process design and optimization [3, 6, 10].

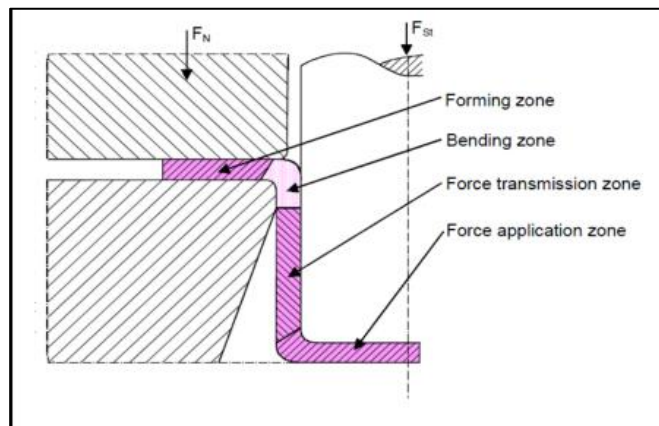


Figure 1. Deformation zones and force distribution in the deep drawing process.

Recent studies emphasize that many numerical deep drawing analyses still rely on nominal CAD geometry, even though real formed components exhibit spring-back distortions, local thinning, and non-uniform wall thickness [5, 9]. These geometric deviations have been shown to significantly influence structural performance and stiffness. Consequently, neglecting the actual post-forming geometry may lead to substantial inaccuracies in subsequent mechanical evaluations, particularly for components subjected to service loads such as internal pressure [9].

This limitation forms the core motivation of the present study. The primary objective is to investigate how the cold-worked state generated during the deep drawing process influences the mechanical behavior of a vessel subjected to internal pressure. To achieve this, an axisymmetric deep drawing simulation is first constructed using PrePoMax, and the numerical solution is obtained using CalculiX [10], an open-source finite element solver capable of handling nonlinear elastoplastic behavior and large deformations. The deep drawing simulation provides the realistic post-forming geometry, including thickness variation, accumulated plastic strain, and spring-back effects.

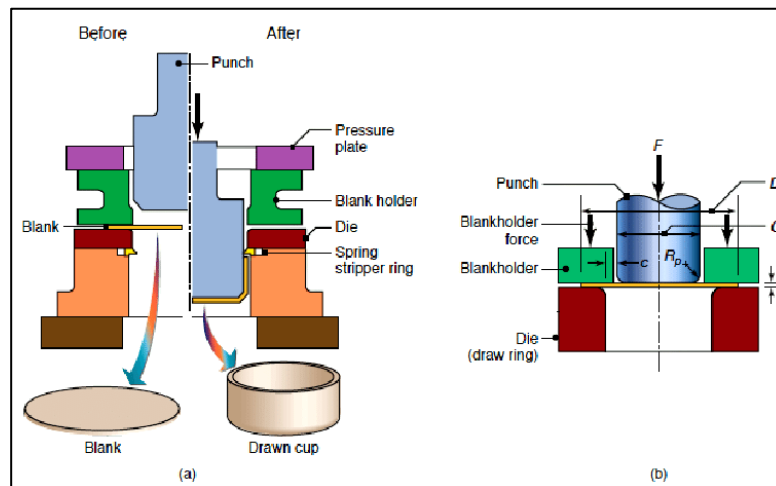


Figure 2. Deep drawing process , (a) drawing a cup, (b) process variables.

In the second stage of the analysis, the deformed geometry obtained from the forming simulation rather than an idealized CAD model is subjected to internal pressure loading to evaluate the true structural response of the cold-worked component [9]. By preserving the complete deformation history through this two-stage FEM workflow, the study aims to deliver more accurate and physically meaningful predictions in deep-drawn vessels.

2. LITERATURE REVIEW

Deep drawing has been extensively studied in the literature due to its wide industrial application and the complex deformation mechanisms involved during the forming process. Early investigations primarily focused on understanding the fundamental mechanics of material flow, thickness variation, and failure modes such as wrinkling and tearing.

2.1. Analytical and Experimental Investigations

Initial studies on deep drawing relied on analytical approaches, including slab methods and slip-line field theory, to estimate drawing forces and limiting drawing ratios. While these methods provided valuable insight into the basic mechanics of the process, their applicability was limited by simplifying assumptions regarding material behavior, friction conditions, and geometry [1]. Experimental investigations later complemented these analytical models by examining the influence of process parameters such as blank holder force, punch radius, die radius, and lubrication conditions on formability and thickness distribution [2].

Several experimental studies demonstrated that deep drawing is highly sensitive to friction and blank holder conditions. Variations in local friction have been shown to significantly affect material flow and localized thinning, particularly in the wall and corner regions of drawn components.

$$\sigma = K\varepsilon^n \quad 2.1$$

where σ is the true stress, ε is the equivalent plastic strain, K is the strength coefficient, and n is the strain-hardening exponent.

$$\tau = \mu \cdot P \quad 2.2$$

where τ is the frictional shear stress, μ is the friction coefficient, and P is the contact pressure.

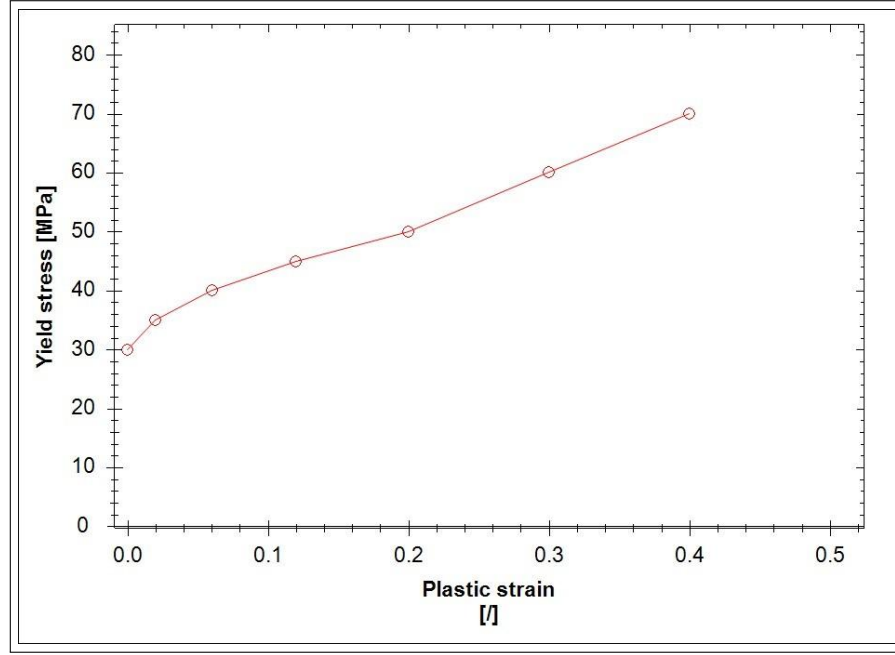


Figure 3. 1060 Aluminum Stress Strain Curve

2.2. FINITE ELEMENT MODELLING OF DEEP DRAWING

With the advancement of computational mechanics, the Finite Element Method (FEM) emerged as a powerful tool for analyzing deep drawing processes. FEM-based studies enabled the realistic simulation of large plastic deformations, complex contact interactions, and nonlinear material behavior. Early finite element applications laid the foundation for numerical metal forming analysis by demonstrating the feasibility of simulating forming forces, strain distributions, and thickness evolution [3]. Subsequent research focused on improving numerical accuracy by incorporating advanced material models, refined contact algorithms, and adaptive meshing strategies. Finite element analyses have been successfully used to predict key forming phenomena such as wrinkling, tearing, ironing, and spring-back [4,5]. These studies highlighted the importance of mesh refinement in regions with high strain gradients, especially around punch radii and die corners.

$$LDR = \frac{D_{blank}}{D_{punch}} \quad 2.2$$

where D_{blank} is the initial blank diameter and D_{punch} is the punch diameter.

2.3. EFFECT OF PROCESS PARAMETERS AND MATERIAL BEHAVIOUR

Numerous studies have investigated the influence of process parameters on deep drawing performance. Parameters such as blank holder force, punch speed, friction coefficient, and tool geometry have been shown to strongly affect the limiting drawing ratio and thickness distribution [2,6]. Material behavior, particularly strain hardening and anisotropy, also plays a critical role in determining formability and failure limits.

Research on pre-straining effects has demonstrated that prior plastic deformation alters the flow and fracture behavior of sheet metals, influencing their response during subsequent forming or loading stages [7]. These findings emphasize the importance of accounting for the cold-working history when evaluating the mechanical performance of formed components.

$$F_{bh} = A.p \quad 2.3$$

where F_{bh} is the blank holder force, A is the contact area of the blank holder, and p is the applied holding pressure.

2.4. SPRING-BACK AND POST-FORMING GEOMETRY

Spring-back remains one of the most challenging aspects of deep drawing simulation. Several studies have focused on predicting spring-back behavior and residual stress distributions after unloading, particularly for large-scale forming applications [5]. Accurate spring-back prediction is essential for obtaining realistic post-forming geometries.

Recent literature increasingly emphasizes that many structural analyses of deep drawn components are performed on idealized CAD geometries, neglecting the geometric deviations introduced during forming. However, studies have shown that local thinning, residual stresses, and spring-back-induced distortions can significantly affect stiffness, strength, and load-carrying capacity [5,8].

2.5. VON MISES YIELD CRITERION AND STRESS THEORY

To accurately predict the onset of plastic deformation in ductile materials, a robust yield criterion is required. In computational mechanics and manufacturing simulations, the Von Mises yield criterion, also known as the maximum distortion energy theory, is the most widely adopted approach. This theory postulates that yielding begins when the distortion energy per unit volume in a material reaches a level equal to the distortion energy at the yield point in a simple tension test.

The yielding criterion for the axisymmetric model is determined by the Von Mises stress, which combines the radial (σ_r), hoop (σ_θ), and axial (σ_z) stress components [10]:

$$\sigma_{eq} = \sqrt{\frac{(\sigma_r - \sigma_\theta)^2 + (\sigma_\theta - \sigma_z)^2 + (\sigma_z - \sigma_r)^2 + 6\tau_{rz}^2}{2}} \quad 2.5$$

3. METHODOLOGY

This study investigates the mechanical effects of cold-working introduced by the deep drawing process using the Finite Element Method (FEM). The numerical methodology is structured as a clear and systematic workflow consisting of three main stages: modelling, meshing, and solution. The procedure is formulated in a general framework using PrePoMax as the pre-processor and CalculiX as the solver, ensuring applicability to sheet metals commonly used in forming operations.

3.1. NUMERICAL MODELLING STRATEGY

The deep drawing process is modelled numerically using a two-dimensional axisymmetric representation to significantly reduce computational costs while preserving physical accuracy in stress distribution. This approach allows for a highly refined mesh in critical zones, such as fillet radii, without the excessive computational burden of 3D models.

The simulation strategy is bifurcated into two interdependent stages:

1. **Manufacturing Phase:** A flat 1060 Aluminum blank is formed into a cup to capture manufacturing-induced effects like thinning and plastic strain accumulation.
2. **Structural Assessment Phase:** The deformed geometry and the state variables, specifically the Equivalent Plastic Strain (PEEQ), are utilized as the initial state for subsequent structural loading to account for the actual work-hardened state of the vessel.

3.2. FINITE ELEMENT FORMULATION AND ELEMENT SELECTION

The selection of the finite element type is a decisive factor in the predictive accuracy of metal forming simulations. For the numerical discretization of the deformable Aluminum 1060 blank, second-order quadratic axisymmetric continuum elements with reduced integration (CAX8R) were implemented. This specific formulation was strategically chosen to address the high degree of geometric and material nonlinearity inherent in the deep drawing process.

3.2.1. CAX: Axisymmetric Continuum Formulation

The CAX designation signifies that the model is formulated as an axisymmetric continuum. Deep drawing of circular cups exhibits rotational symmetry, which allows the reduction of a complex 3D problem into a 2D cross-sectional analysis.

Computational Efficiency: This approach significantly reduces the degrees of freedom (DOF), leading to enhanced computational efficiency without sacrificing physical accuracy.

Hoop Stress Representation: Unlike standard plane stress or plane strain elements, the CAX formulation explicitly accounts for circumferential (hoop) stresses. These stresses are the primary drivers for flange thickening and potential wrinkling, making the CAX formulation indispensable for a realistic representation of the forming mechanics.

Table 1. Material Properties

Property	Symbol	S185 Steel	1060 Aluminum
Young's Modulus	E [MPa]	210000	69000
Poisson's Ratio	ν	0.28	0.33
Density	ρ	7.80E-09	2.70E-09
Initial Yield Stress	σ_y [MPa]	N/A (Rigid)	30

3.2.2.8: Higher-Order (Quadratic) Interpolation

The digit 8 denotes that the element is an eight-node biquadratic element. In contrast to 4-node linear elements, quadratic elements include mid-side nodes, which offer several critical advantages:

- **Geometry Conformity:** The presence of mid-side nodes allows the elements to conform smoothly to the curved surfaces of the punch nose ($R_f P$) and die shoulder ($R_f D$). This prevents the "faceted" contact typical of linear elements, which can lead to artificial pressure spikes and numerical instability.
- **Bending Accuracy:** Deep drawing involves continuous bending and unbending as the sheet flows over the die radius. Quadratic elements are far superior at capturing

the steep stress and strain gradients across the thickness of the blank with fewer layers than linear elements would require.

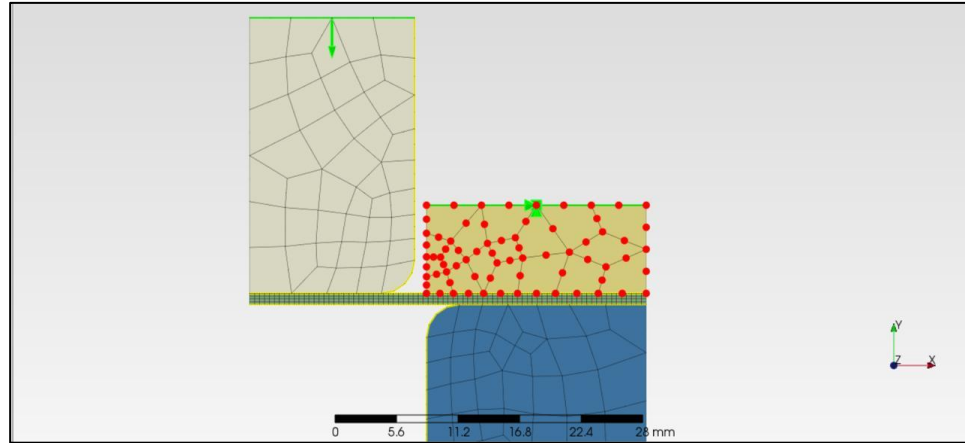


Figure 4. CAX8R

3.2.3.R: Reduced Integration Technique

The suffix R indicates that the element utilizes a reduced integration scheme, which is a strategic numerical choice for large-strain plasticity simulations.

- **Prevention of Volumetric Locking:** During the plastic phase, the Aluminum 1060 alloy deforms isochorically (maintaining constant volume). Fully integrated elements often suffer from volumetric locking, where the element becomes artificially stiff. Reduced integration effectively bypasses this numerical constraint, allowing for a more realistic prediction of the material flow.
- **Shear Locking Mitigation:** It also mitigates shear locking, which occurs when elements are subjected to bending, ensuring that the predicted thinning at the punch radius is governed by physics rather than numerical errors.

3.2.4. Synthesis of the High-Fidelity Selection

The adoption of the CAX8R formulation represents a high-fidelity modeling strategy designed to capture the complex manufacturing history of the component. By combining the geometric flexibility of higher-order nodes with the numerical stability of reduced integration, the simulation is capable of accurately predicting precise thickness variations and residual stress states of the Aluminum 1060 cup. This configuration ensures that the

open-source integration of Gmsh, CalculiX, and PrePoMax operates at a level of accuracy comparable to high-end commercial FEA suites.

3.3. SOLUTION PROCEDURE AND STEP DEFINITIONS

The numerical solution was executed in four sequential steps to realistically replicate the physical manufacturing stages and to investigate the influence of varying boundary conditions on material flow. Unlike conventional single-stroke deep drawing, this multi-step approach isolates the effects of blank holder pressure and subsequent release on the thinning and hardening characteristics of the specimen.

Table 2. Geometrical Parameters

Parameter	Symbol	Value [mm]
Initial Blank Thickness	T_0	1
Blank Outer Radius	R_{blank}	36,1
Punch Width	W_{punch}	15
Punch Fillet Radius	R_p	3
Die Fillet Radius	R_d	3
Radial Clearance	c	1,1

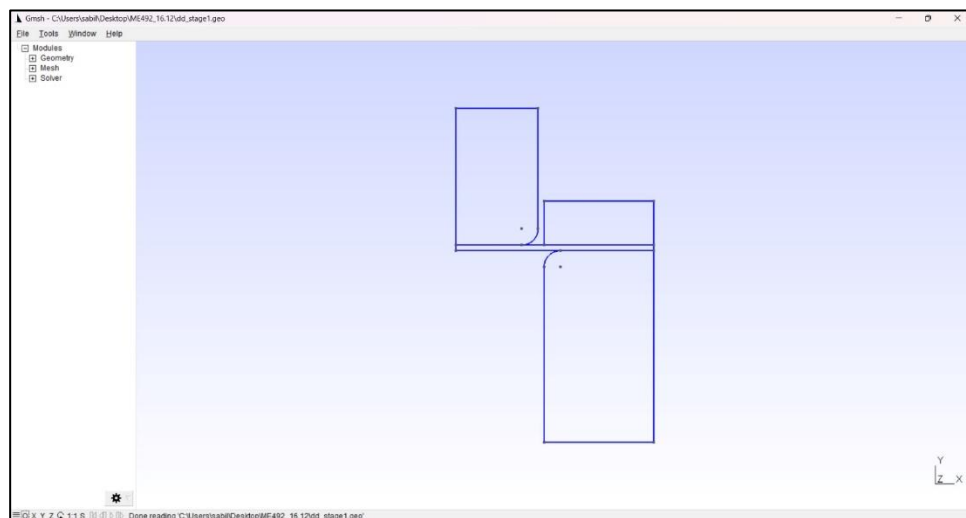


Figure 5. Initial 2D axisymmetric geometric model of the deep drawing setup (punch, die, holder, and blank) created in Gmsh, prior to mesh discretization.

3.3.1. Step 1: Clamping and Initial Forming (0–5 mm)

The primary stage of the simulation is dedicated to establishing a stable deformation environment by applying a controlled clamping pressure to the flange area. This initial constraint is crucial for mitigating the risk of wrinkling a common defect in deep drawing while simultaneously regulating the rate of material flow into the die cavity.

During this phase, the punch initiates its downward stroke, reaching a displacement of 5 mm. As the punch makes contact with the Aluminum 1060 blank, the material begins to conform to the punch nose geometry, marking the transition from elastic to plastic behavior. This step is instrumental in defining the initial cup base and triggering the first wave of plastic strain accumulation, particularly at the punch radius. The mechanical response observed in this interval sets the boundary conditions for the hardening characteristics of the alloy, establishing the foundation for the subsequent stages of the forming process.

3.3.2. Step 2: Boundary Condition Shift (Holder Retraction)

Following the initial 5 mm stroke, the blank holder is translated 3 mm upward while the punch maintains its current position. This intermediate step is a critical aspect of the methodology, as it abruptly reduces the compressive constraint and the friction-induced resistance on the flange. This phase allows for the observation of the material's mechanical response to the release of the clamping force and evaluates the stress relaxation within the flange region.

3.3.3. Step 3: Final Forming in Unconstrained State (5–10 mm)

With the blank holder maintained in its elevated (inactive) position, the punch resumes its downward movement for an additional 5 mm, reaching a final depth of 10 mm. This phase simulates a "free-forming" or low-pressure drawing state. By continuing the draw without holder pressure, the simulation isolates the evolution of thinning at the punch radius and thickening at the flange under reduced radial restriction.

3.3.4. Step 4: Unloading and Spring-back Identification

The final stage involves the full retraction of the punch to its home position, effectively removing all external tool contacts. This unloading step is essential to distinguish the permanent plastic deformation from the total elastic-plastic displacement. The resulting geometry provides the "as-manufactured" state, including the final thickness profile and the manufacturing-induced residual stresses.

3.4. NUMERICAL FORMULATION AND COUPLING

The forming stages were solved using an implicit time integration scheme, which provides high accuracy for quasi-static processes involving complex contact and large plastic deformations. The cold-working history of the material was continuously updated based on equivalent plastic strain (PEEQ) accumulation. The structural evaluation under internal pressure, which constitutes the second phase of this research, utilizes the deformed and hardened geometry from Step 4 as its initial state. This coupling ensures that the structural performance assessment fully accounts for the thickness variations and strength increases localized during the manufacturing process.

Table 3. Summary of the displacement-controlled operational steps.

Step	Operational Description	Punch Disp. [mm]	Holder Disp. [mm]
Step 1	Initial Drawing (Constrained)	-5.0	0.0
Step 2	Holder Retraction	-5.0 (Hold)	+3.0
Step 3	Final Drawing (Unconstrained)	-10.0	+3.0
Step 4	Final Retraction	0.0	+3.0

4. ANALYSIS AND RESULTS

The results of the axisymmetric deep drawing simulation are evaluated in this section through a systematic analysis of discretization quality, displacement fields, stress distributions, and reaction forces. The primary objective is to validate the numerical model and observe the plastic flow behavior of the 1060 Aluminum alloy under the specified industrial forming conditions.

4.1. VERIFICATION OF MANUAL ELEMENT TRANSFORMATION (CAX8R)

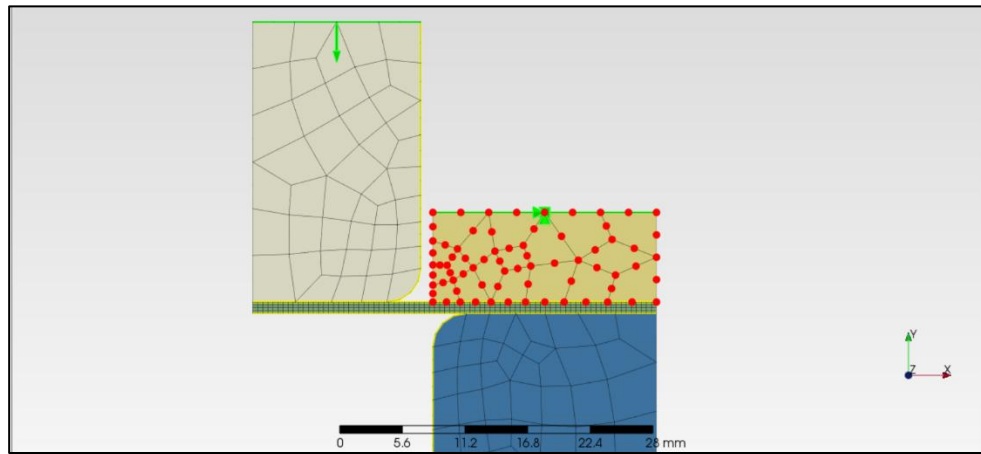


Figure 6. Quadratic node distribution on the blank holder assembly.

The discretization of the blank holder was verified by visualizing the mid-side nodes, indicated by red markers in Figure 7. The presence of these nodes confirms the transition to second-order biquadratic elements, which are essential for maintaining contact stability and ensuring that the restraining forces are accurately distributed across the flange region during the forming increments.

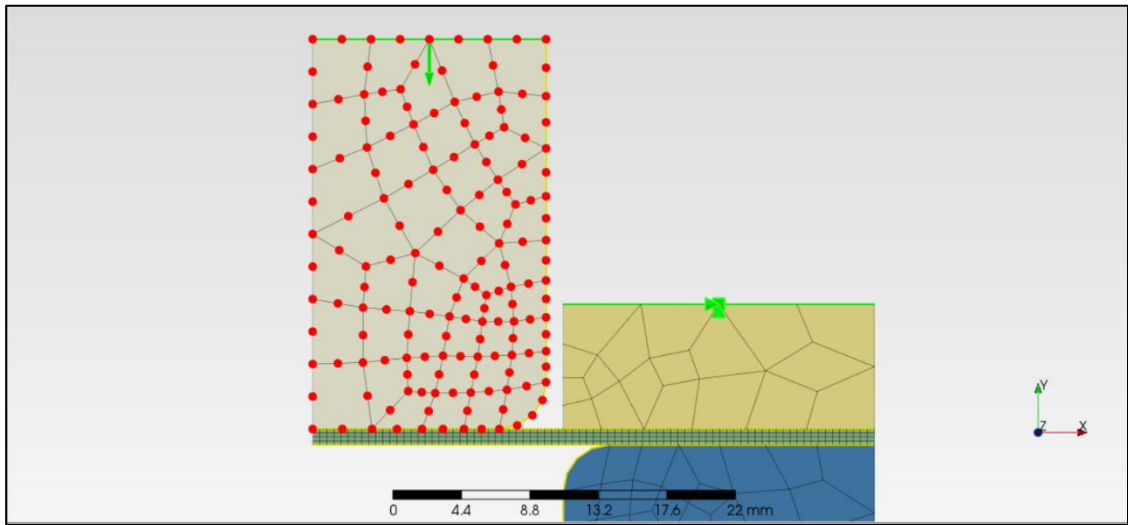


Figure 7. Quadratic node distribution on the punch assembly.

As illustrated in Figure 8, the punch mesh includes mid-side nodes to provide a smoother representation of the contact interface. This refined configuration is critical for modeling the bending mechanics at the cup bottom accurately and prevents artificial stiffening (shear locking) as the punch reaches its maximum stroke.

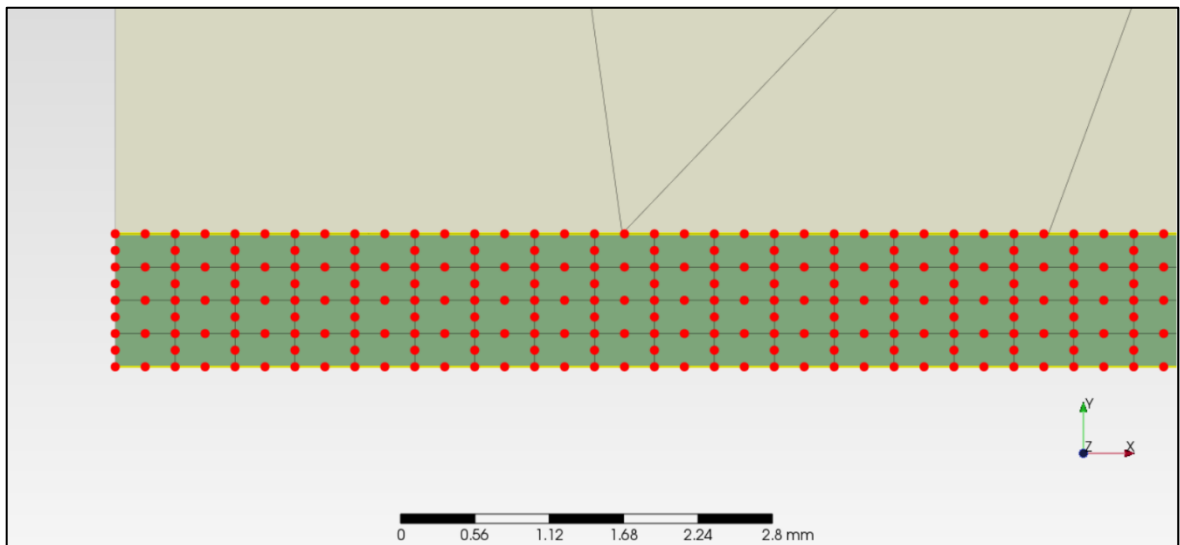


Figure 8. Close-up view of the quadratic node distribution on the 1060 Aluminum blank.

The aluminum blank's mesh architecture, presented in Figure 9, utilizes a structured quadrilateral grid with 4 element layers through its thickness. The implementation of mid-side nodes ensures that hoop stresses and thickness variations (thinning) are calculated with quadratic accuracy, providing a reliable foundation for the subsequent structural assessment phase.

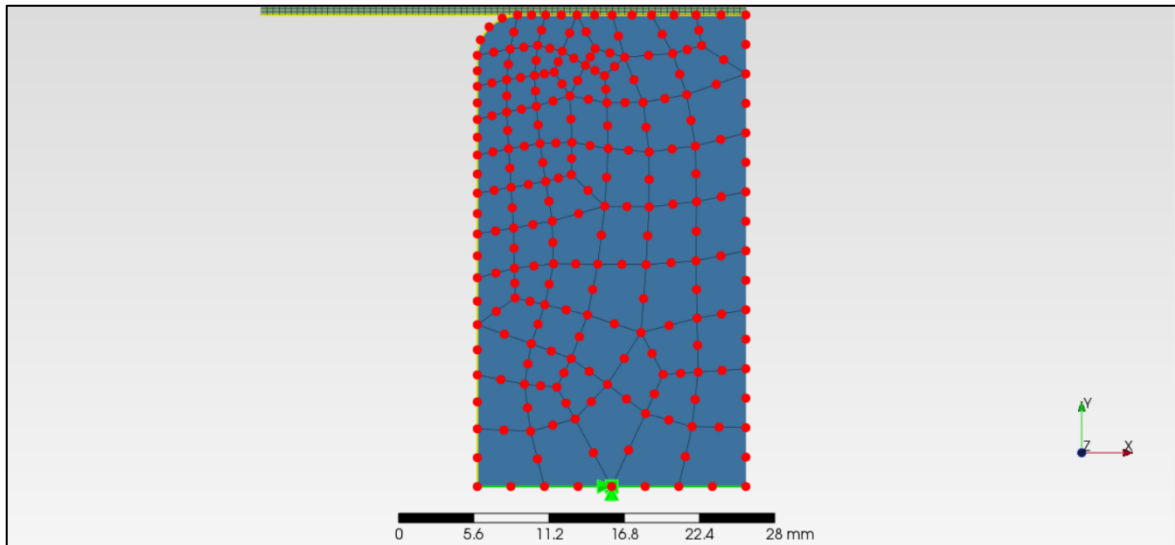


Figure 9. Quadratic node distribution on the die assembly.

The die shoulder discretization is shown in Figure 10, where the quadratic nodes enable precise tracking of the aluminum blank as it flows into the die cavity. This high-resolution mesh at the die radius is necessary to capture the complex bending and unbending stresses without numerical diffusion.

4.2. EQUIVALENT TOTAL STRAIN ANALYSIS

The distribution of Equivalent Total Strain was analyzed to quantify the cumulative deformation experienced by the Aluminum 1060 blank throughout the forming process. This parameter represents the numerical summation of both elastic and plastic strain components. The analysis of the five critical frames is as follows:

Onset of Deformation : In the initial stage of tool engagement, total strain concentrations begin to emerge at the punch nose radius. At this point, the strain is predominantly within the elastic-plastic transition zone, as the material starts conforming to the tool geometry.

Completion of Step 1 (5 mm Stroke): Upon reaching the 5 mm downward displacement, a clear strain gradient is established. The total strain values at the punch radius reflect significant plastic flow, indicating that the material has undergone substantial cold-working in this region.

Blank Holder Retraction Response (Step 2): Following the 3 mm upward translation of the blank holder, the total strain distribution shows localized stability. While the plastic

component remains constant as the punch is held, the analysis identifies a minor redistribution of the total strain in the flange region due to the removal of the clamping constraint.

Peak Strain at Final Depth (10 mm Stroke): At the maximum punch stroke of 10 mm, the equivalent total strain reaches its peak magnitude. The highest strain concentrations are observed at the punch radius and the cup's lower wall, highlighting the zones of maximum material stretching and hardening during the unconstrained forming phase.

Post-Unloading Final State (Step 4): After the full retraction of the tools, the total strain values exhibit a slight reduction. This change corresponds to the recovery of the elastic strain component, representing the spring-back effect. The remaining strain at this stage defines the permanent "as-manufactured" geometry of the Aluminum 1060 component.

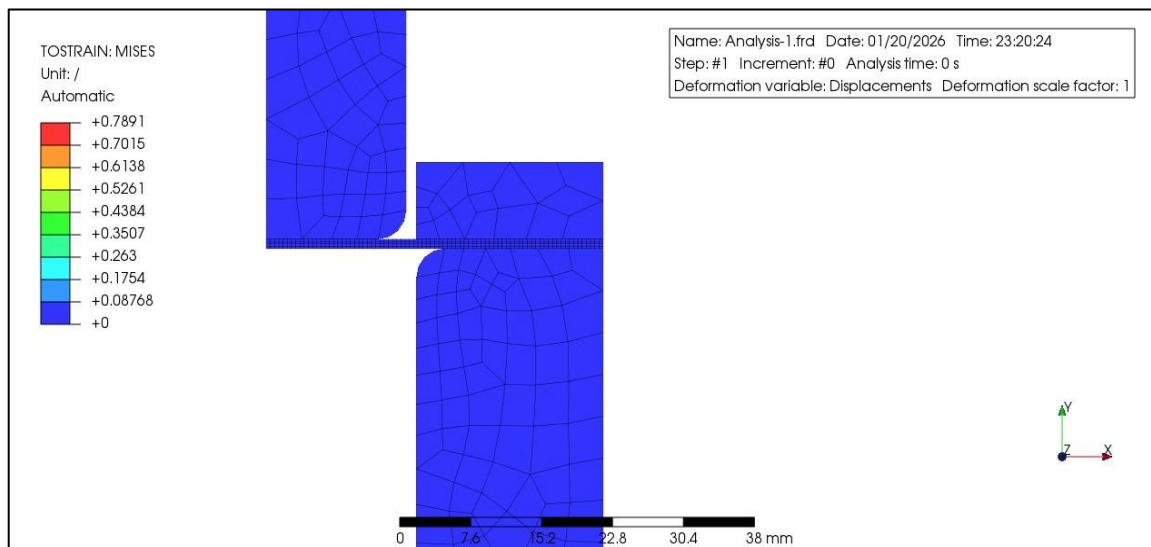


Figure 10. Total Strain Step 1

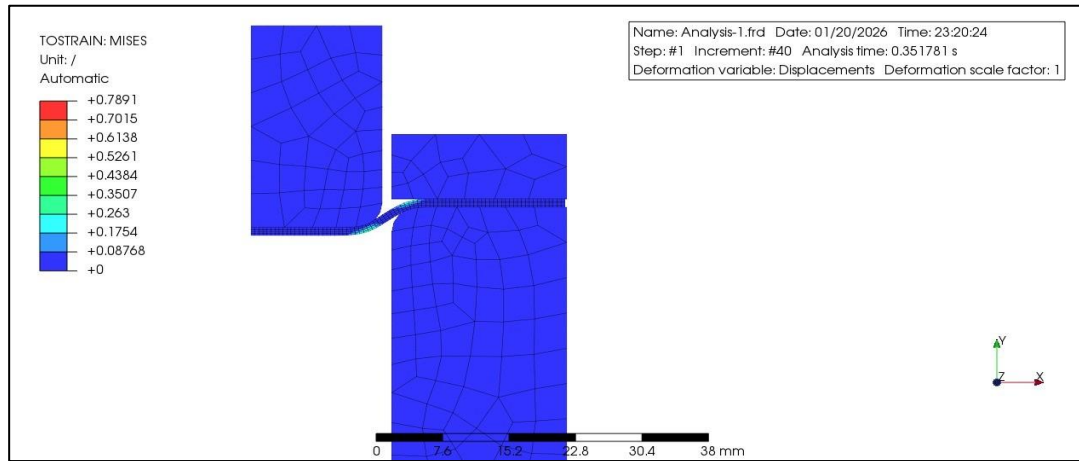


Figure 11. Total Strain Step 2

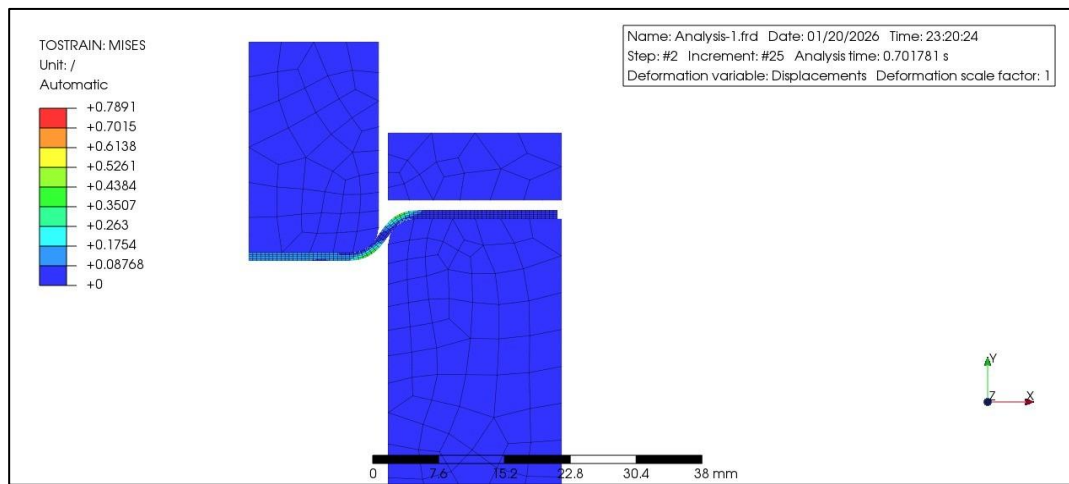


Figure 12. Total Strain Step 3

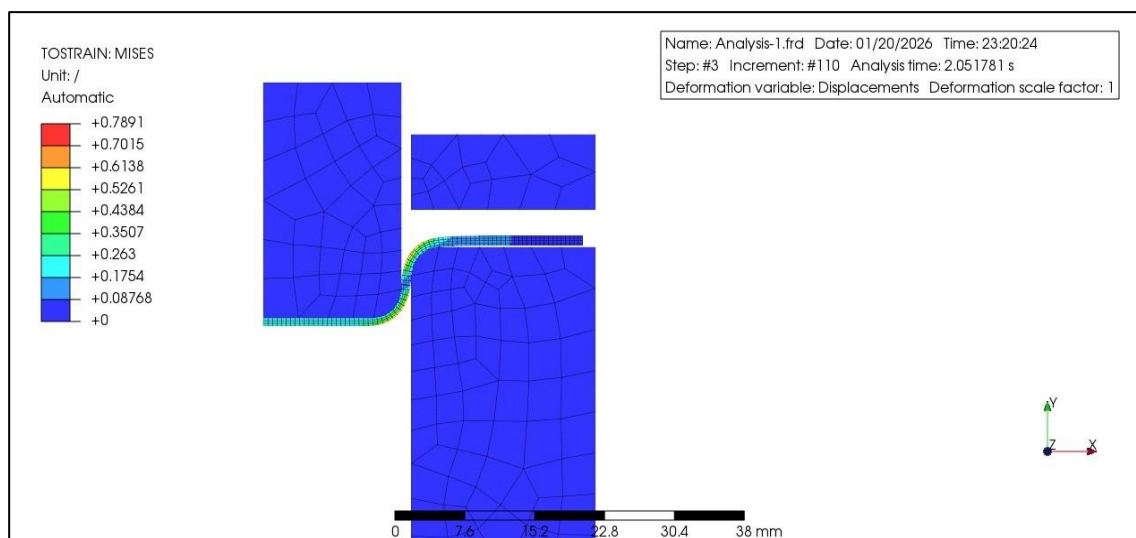


Figure 13. Total Strain Step 4

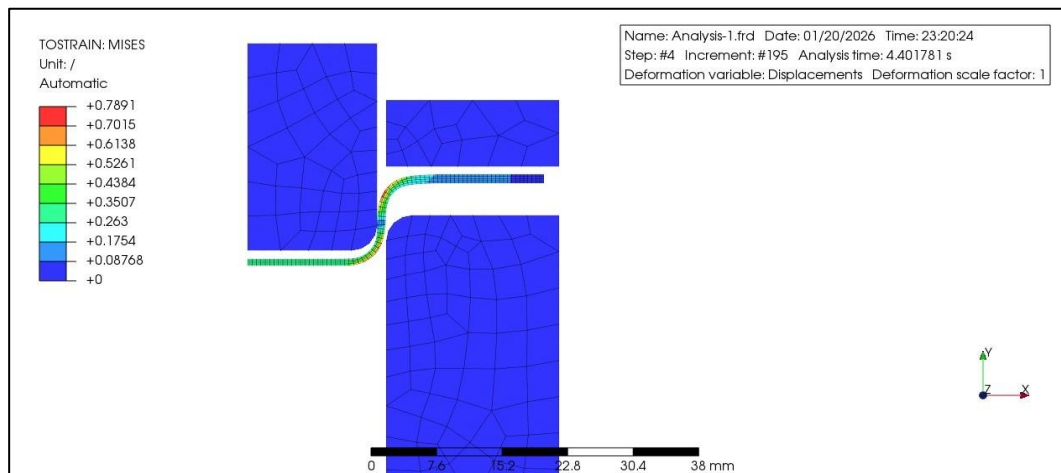


Figure 14. Total Strain Step 5

4.3. STRESS ANALYSIS (VON MISES STRESS)

The Von Mises equivalent stress distribution was monitored to analyze the mechanical response of the Aluminum 1060 blank during the transition from the initial elastic state to the fully developed plastic flow. The following analysis characterizes the stress evolution based on five critical frames extracted from the simulation:

- **Initial Contact:** At the beginning of the simulation, stress concentrations are identified at the punch nose radius. The recorded stress levels are approximately 30 MPa, which is consistent with the initial yield strength of the material, marking the onset of the plastic zone development.
- **Primary Forming Stage:** Upon reaching a punch stroke of 5 mm, the Von Mises stress increases significantly at the primary bending zones. The distribution shows that the material surrounding the punch radius has transitioned into the plastic regime, with stress values exceeding the initial 30 MPa threshold as work-hardening initiates.
- **Holder Retraction Effect (Step 2):** Following the 3 mm upward translation of the blank holder, the analysis reveals a localized reduction in Von Mises stress within the flange region. This stress relaxation is a direct numerical result of removing the clamping pressure, verifying the successful transition to an unconstrained flow state.
- **Peak Deformation State (Step 3):** At the final punch depth of 10 mm, the stress field extends throughout the cup wall. The peak Von Mises stress is observed at the punch and die radii, reaching the saturation limit of approximately 70 MPa,

indicating that these regions have undergone maximum work-hardening during the drawing process.

- **Residual Stress Identification (Step 4):** After the full retraction of the punch, the external mechanical loads are removed. The analysis of this state confirms that the stress does not return to zero; instead, a manufacturing-induced residual stress field remains trapped at the bending zones. These stresses represent the "as-manufactured" internal state of the component.

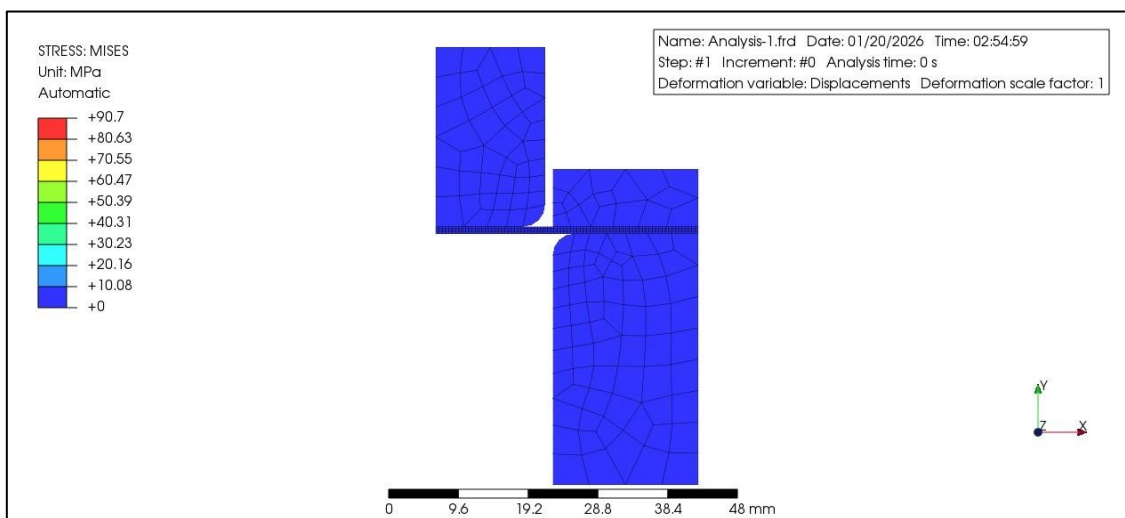


Figure 15. Von Mises Step 1

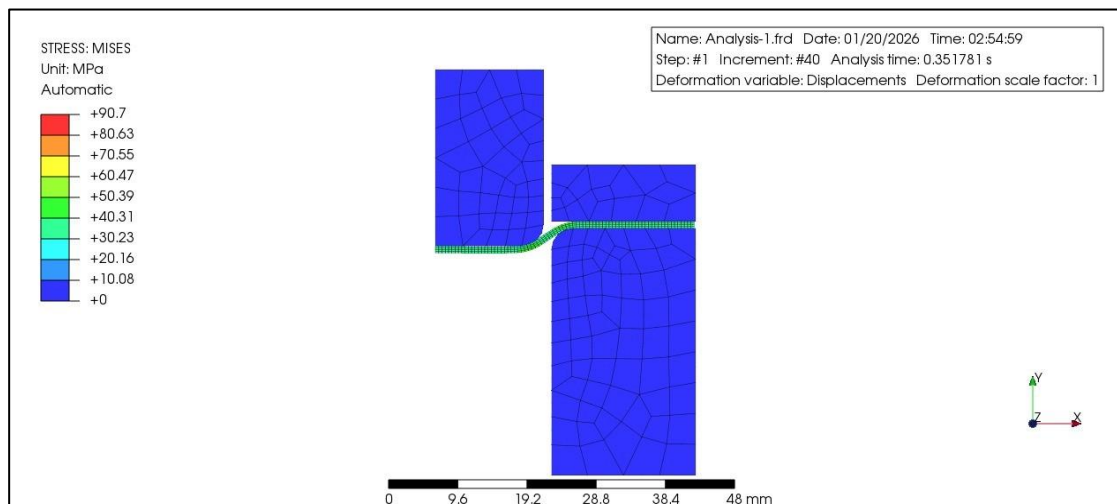


Figure 16. Von Mises Step 2

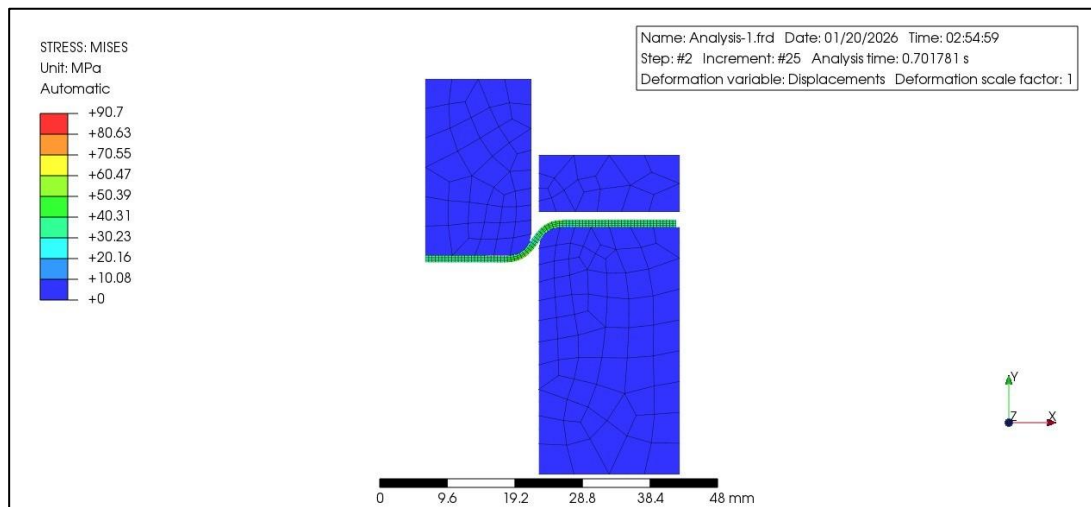


Figure 17. Von Mises Step 3

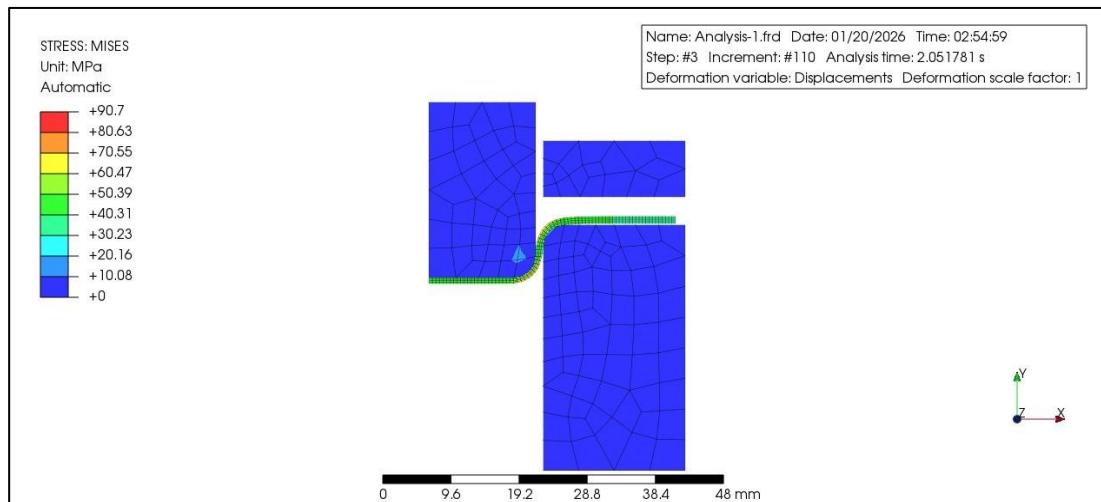


Figure 18. Von Mises Step 4

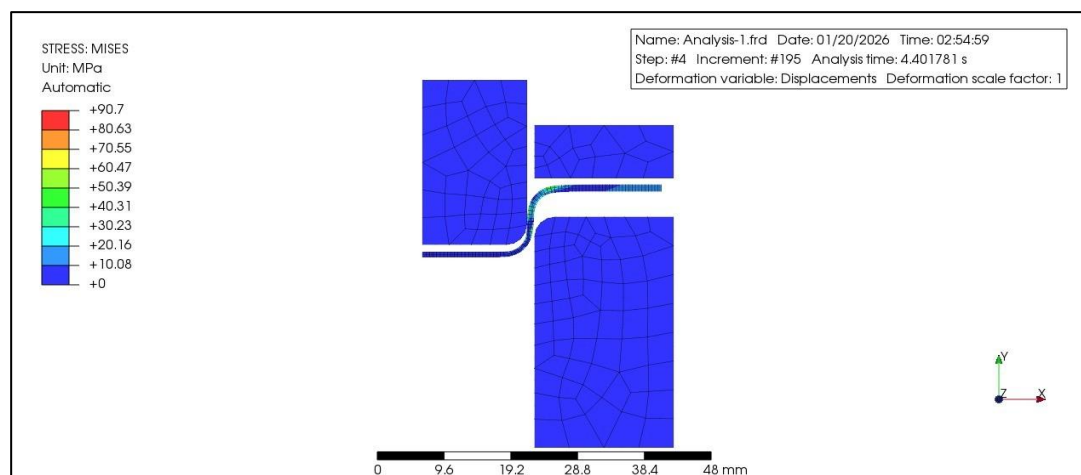


Figure 19. Von Mises Step 5

4.4. EQUIVALENT PLASTIC STRAIN ANALYSIS (PEEQ)

The Equivalent Plastic Strain (PEEQ) analysis was conducted to identify the zones of permanent deformation and localized work-hardening. Unlike total strain, PEEQ specifically represents the irreversible micro-structural changes that occur within the material. In the initial stage (Step 1), as the stress levels surpass the 30 MPa yield threshold of the Aluminum 1060 alloy, plastic strain begins to accumulate at the punch radius. This confirms the onset of the permanent shaping phase, where the blank begins to conform to the punch nose geometry. During the blank holder retraction phase (Step 2), the PEEQ distribution remains perfectly constant. This numerical result verifies that removing the clamping force while the punch displacement is stationary does not induce further plastic flow, highlighting the stability of the existing plastic state. The peak magnitudes of permanent deformation are recorded upon the completion of the final drawing stroke (Step 3). The highest PEEQ values are concentrated at the punch radius and the lower cup wall, reaching the upper limits of the plastic strain defined in the constitutive material model. These regions represent the most heavily cold-worked areas of the finished component. In the final unloading stage (Step 4), following the full retraction of the tools, the PEEQ distribution remains identical to the state observed in Step 3. This behavior confirms the irreversible nature of plastic deformation, as no plastic recovery occurs during the elastic unloading of the material. The final PEEQ map serves as the definitive mechanical "memory" of the manufacturing process stored within the cup.

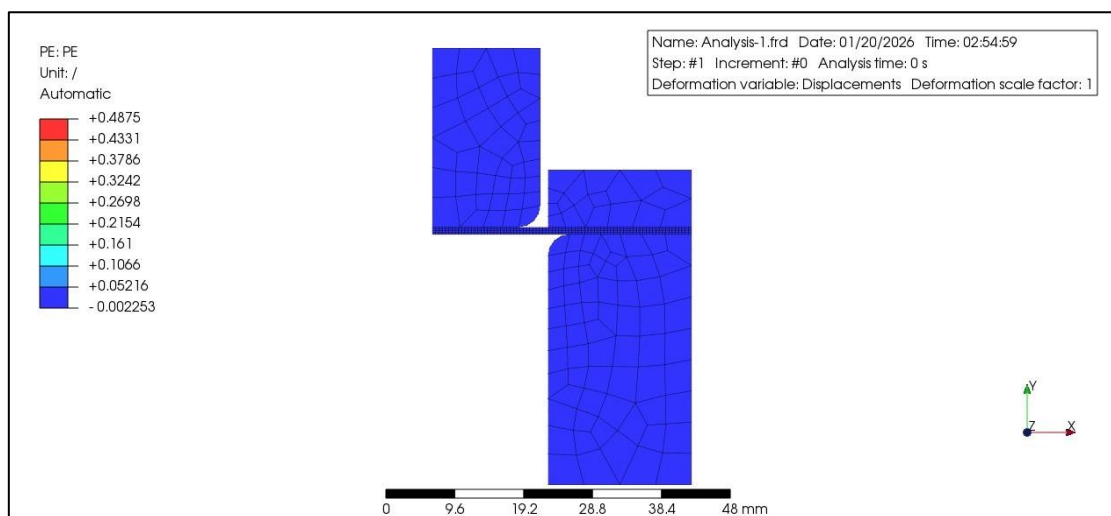


Figure 20. Step 1 PEEQ

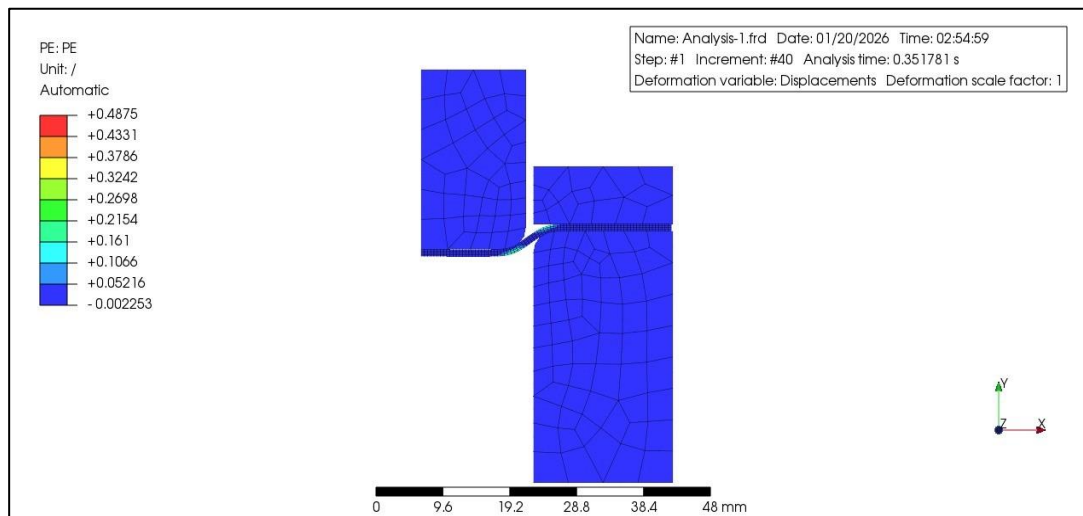


Figure 21. Step 2 PEEQ

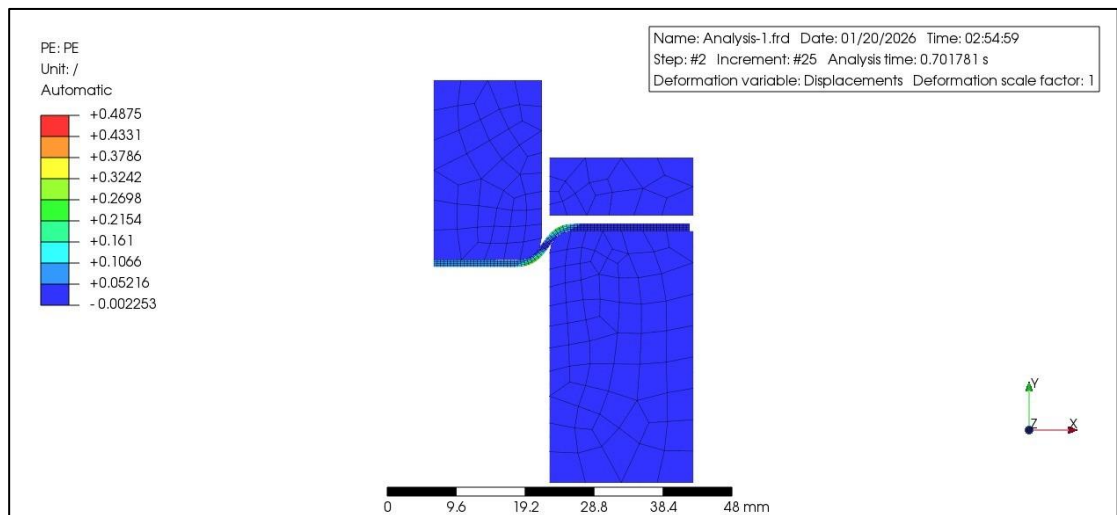


Figure 22. Step 3 PEEQ

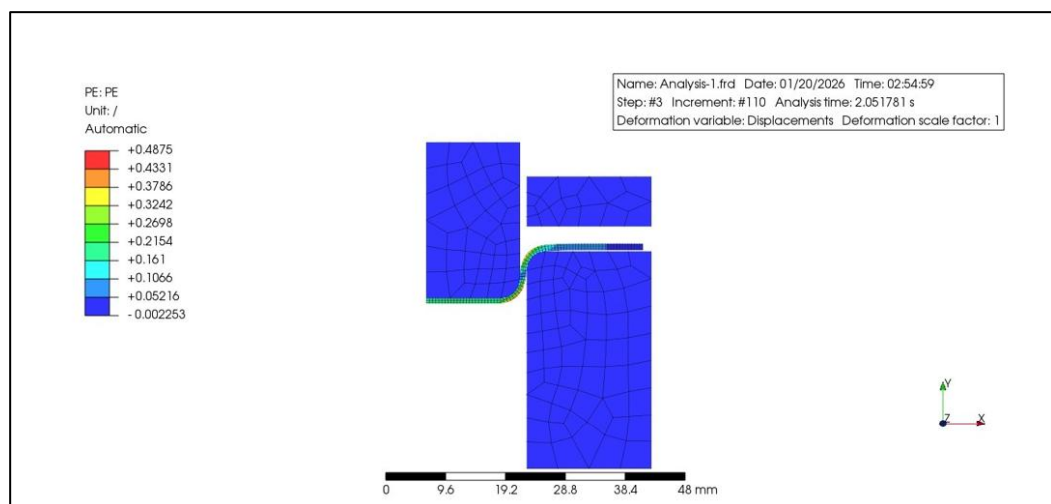


Figure 23. Step 4 PEEQ

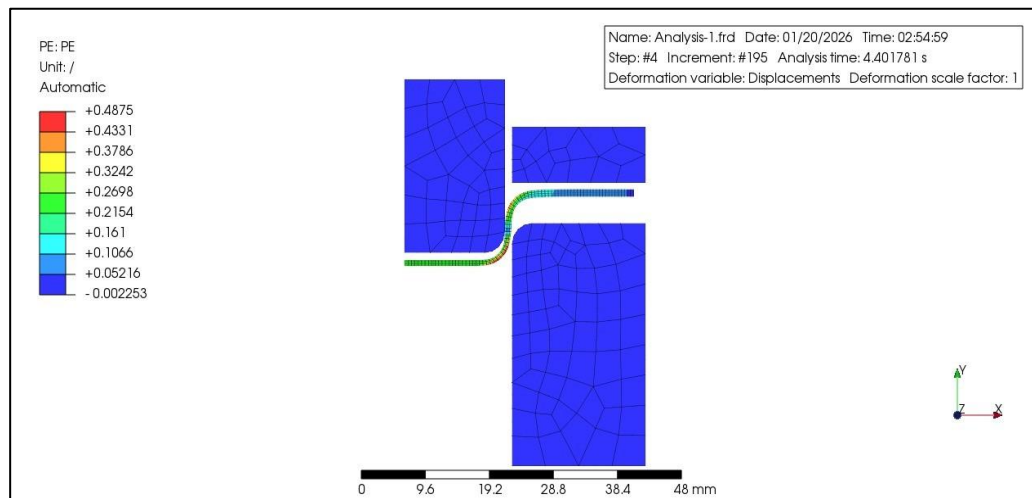


Figure 24. Step 5 PEEQ

4.5. ELASTIC STRAIN ENERGY ANALYSIS

The distribution of Elastic Strain Energy was analyzed to evaluate the potential energy stored within the Aluminum 1060 blank during the deformation process. This energy component is of particular importance as it serves as the primary physical driver behind the spring-back effect observed following tool retraction. The analysis across five critical frames provides a comprehensive look at the energetic state of the component.

During the initial engagement, elastic strain energy begins to accumulate at the contact interfaces, specifically at the punch nose radius. This localized storage represents the material's initial elastic resistance to the downward movement of the punch. As the process progresses to the 5 mm stroke mark, the energy density increases significantly within the primary bending zones. At this stage, the stored energy reaches levels corresponding to the maximum elastic limit of the material just before significant plastic dissipation begins to dominate.

A critical phase of the analysis occurs during Step 2, where the blank holder undergoes a 3 mm upward translation. This maneuver results in a noticeable reduction in elastic strain energy within the flange region. This energy redistribution confirms that the removal of the clamping force allows the material to release a portion of its stored elastic potential, which correlates directly with the stress relaxation phenomena observed in previous sections.

At the final punch stroke of 10 mm, the elastic strain energy reaches its peak global magnitude. The highest energy concentrations are localized at the punch and die radii, representing the maximum "spring-back potential" stored within the cup geometry immediately before the tools are removed.

In the final stage of the simulation (Step 4), upon full retraction of the tools, a substantial decrease in the total elastic strain energy is identified. This reduction effectively visualizes the energy released during the elastic recovery phase, or spring-back. The remaining residual elastic energy indicates that the component has reached a new mechanical equilibrium state in its final, "as-manufactured" configuration.

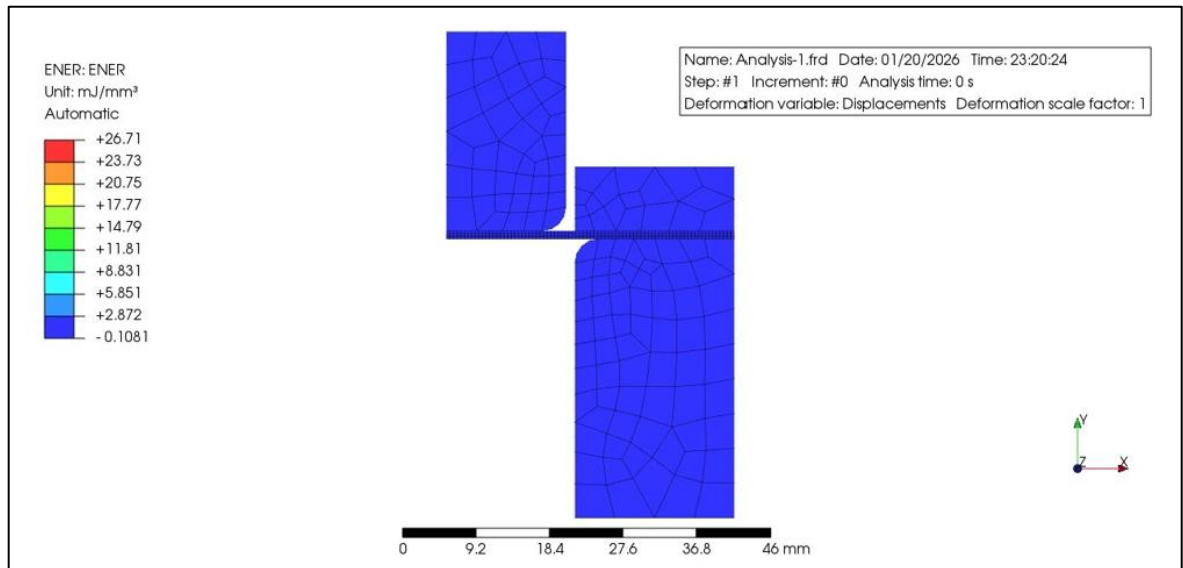


Figure 25. Step 1 Elastic Strain Energy Analysis

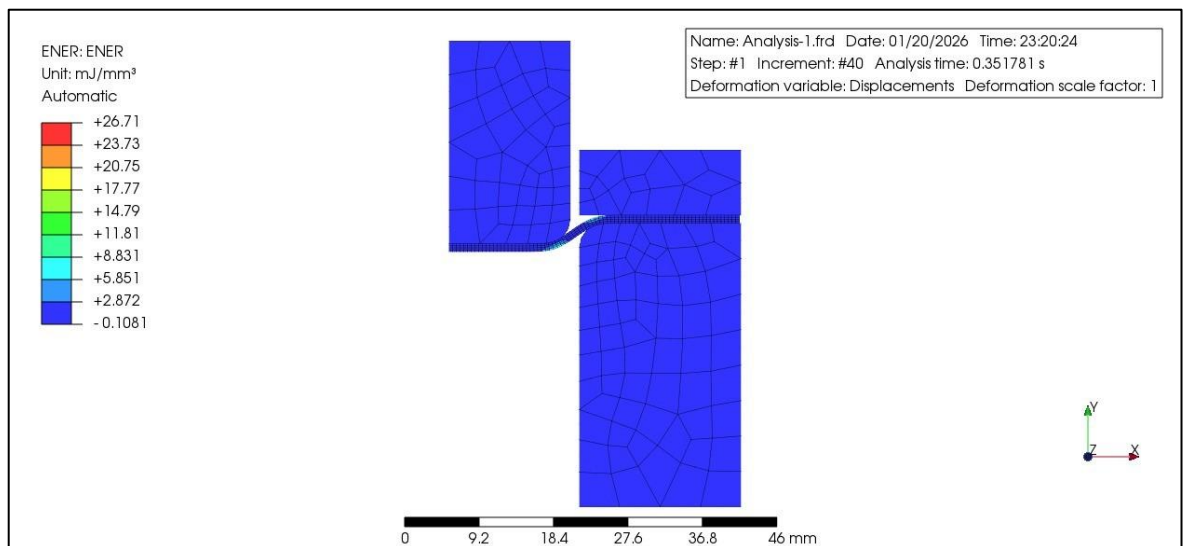


Figure 26. Step 2 Elastic Strain Energy Analysis

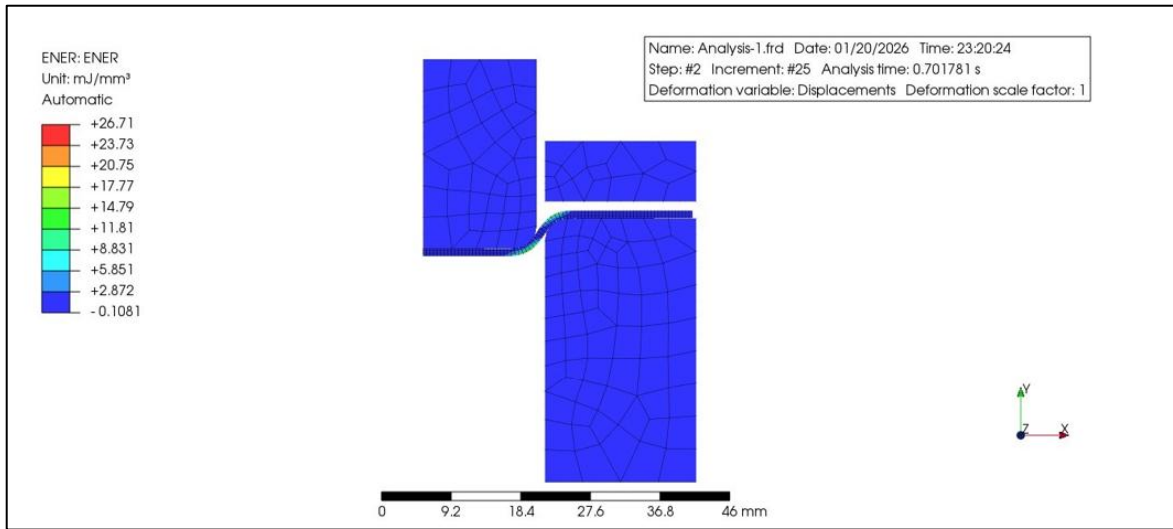


Figure 27. Step 3 Elastic Strain Energy Analysis

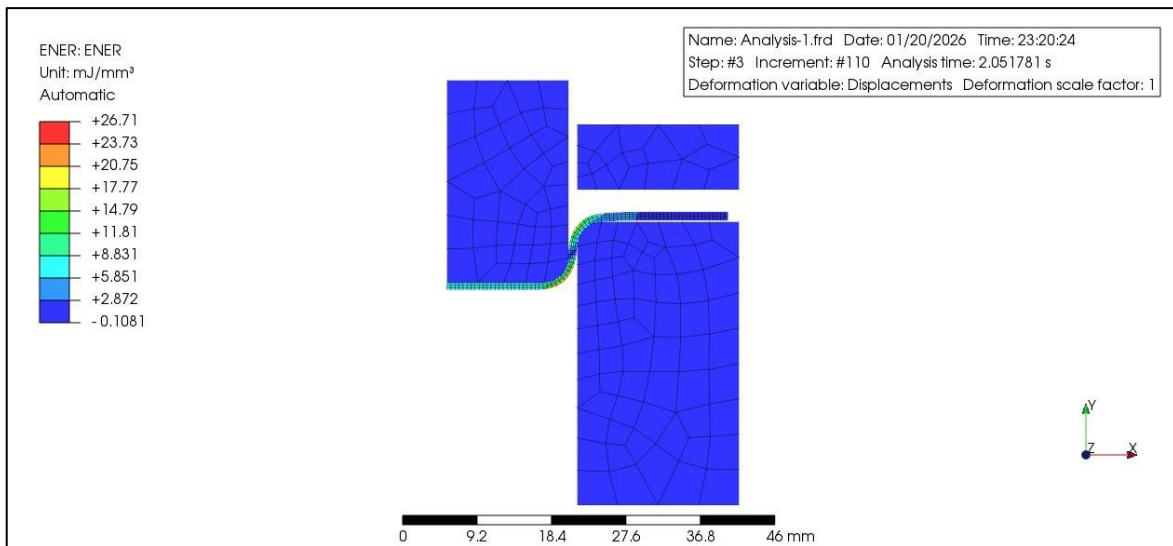


Figure 28. Step 4 Elastic Strain Energy Analysis

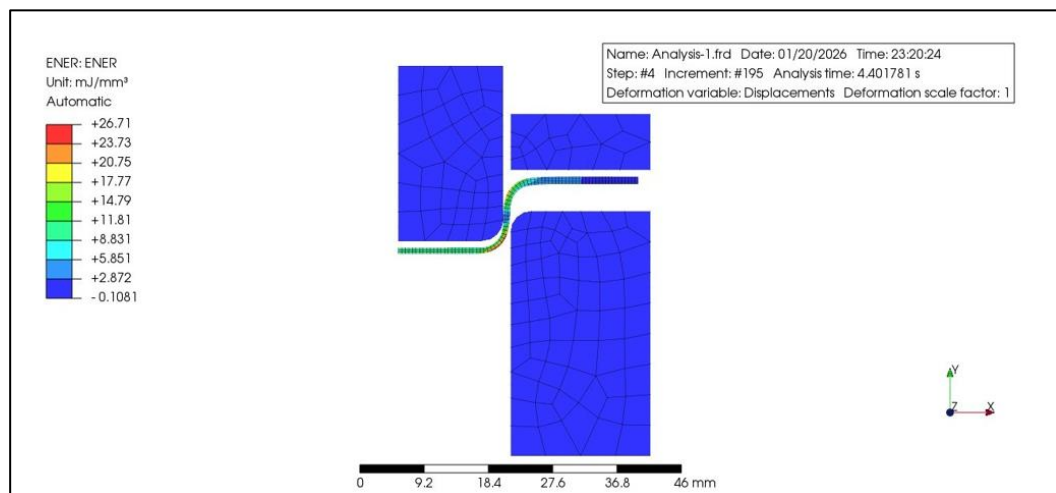


Figure 29. Step 5 Elastic Strain Energy Analysis

4.6. DISPLACEMENT ANALYSIS

The vertical displacement distribution was analyzed to verify the kinematic accuracy of the simulation and to monitor the geometrical evolution of the Aluminum 1060 blank throughout the process. This analysis tracks the progression of the punch stroke and the subsequent blank deformation across five critical frames, ensuring that the numerical model adheres to the defined boundary conditions.

At the beginning of the process, the displacement field confirms the localized initial contact between the punch and the blank. The vertical movement is concentrated at the center of the sheet, marking the initiation of the transition from a flat, two-dimensional blank to a three-dimensional cup geometry. By the conclusion of Step 1, corresponding to a 5 mm stroke, the displacement contours show a maximum value at the punch-blank interface. This distribution verifies that the sheet successfully conforms to the punch movement while the outer flange remains securely constrained by the blank holder.

A unique aspect of this simulation is the boundary condition transition in Step 2. During this blank holder retraction phase, the punch is maintained at a constant vertical displacement of 5 mm. The displacement map at this stage confirms the stability of the punch position while visualizing the 3 mm upward translation of the holder. This specific maneuver is critical as it prepares the blank for unconstrained material flow into the die during the final draw.

Upon reaching the target depth in Step 3, the maximum vertical displacement of 10 mm is recorded at the cup base. The contours at this stage illustrate the complete downward flow of the material, representing the maximum geometric deformation before the unloading phase begins.

Finally, in Step 4, after the tools are retracted to their home positions, the displacement distribution reveals the permanent shape of the cup. Due to the elastic recovery commonly referred to as spring-back the final vertical displacement is slightly lower than the peak 10 mm stroke. This final state accurately represents the "as-manufactured" height of the component, providing a realistic assessment of the dimensional outcome of the forming process.

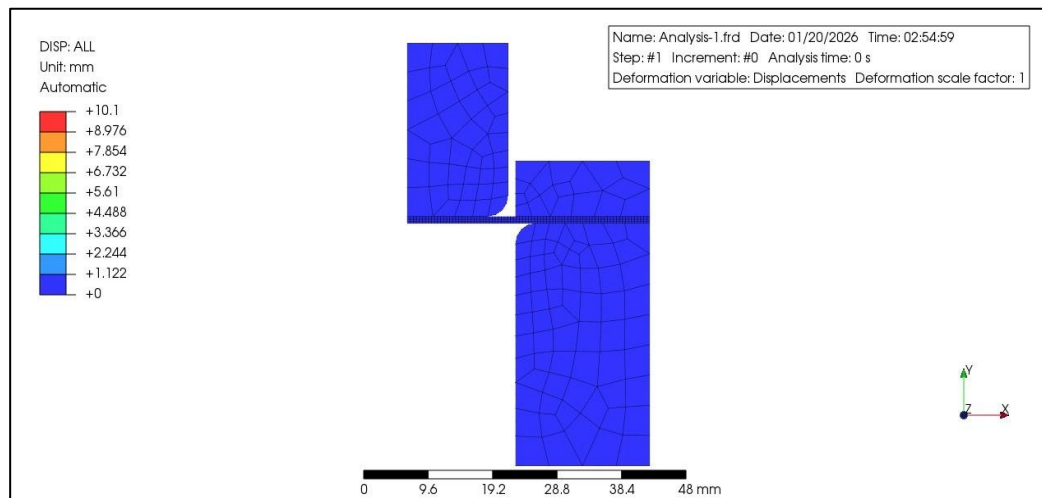


Figure 30. Step 1 Displacement Analysis

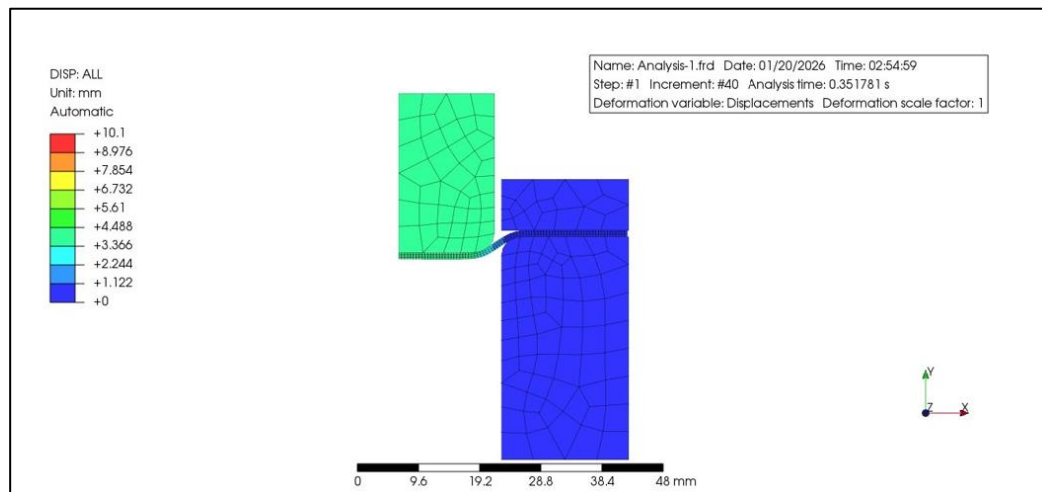


Figure 31. Step 2 Displacement Analysis

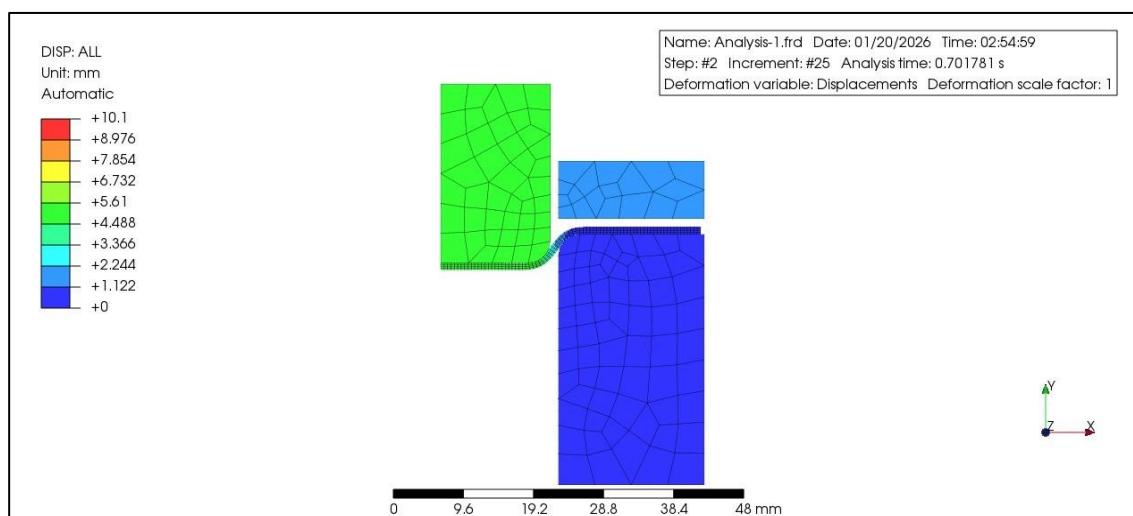


Figure 32. Step 3 Displacement Analysis

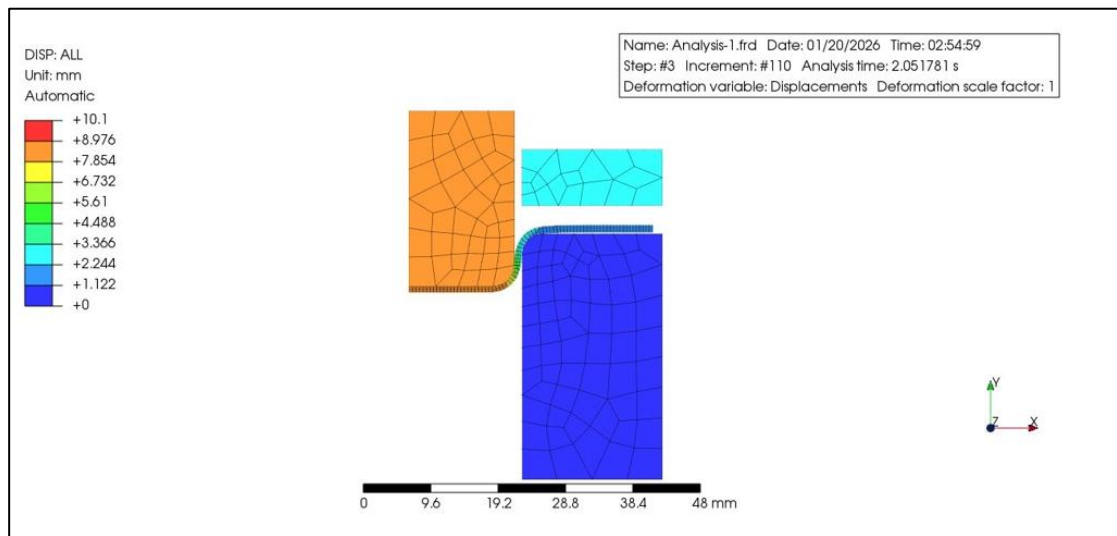


Figure 33. Step 4 Displacement Analysis

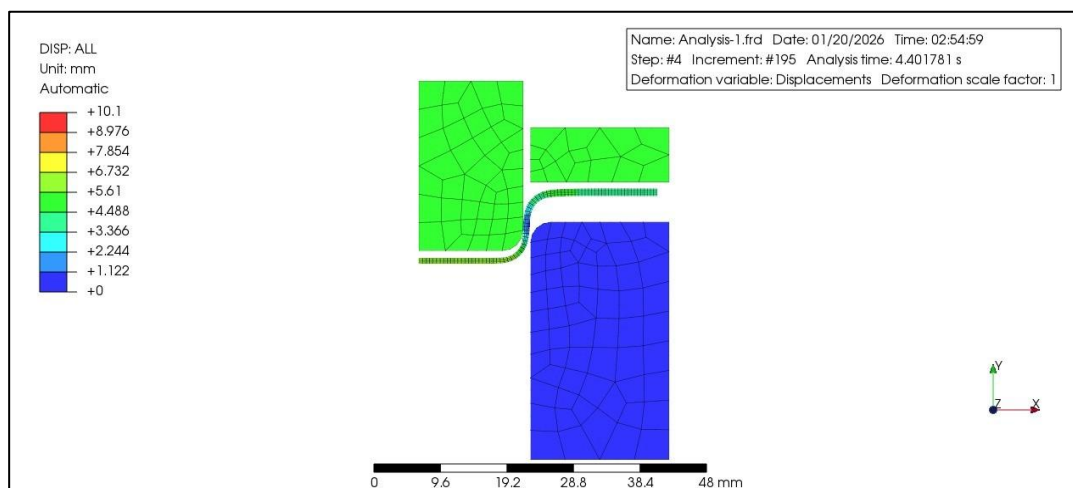


Figure 34. Step 5 Displacement Analysis

4.7. CONTACT PRESSURE ANALYSIS

The distribution of contact pressure was analyzed to evaluate the interaction intensity between the rigid tools (punch, die, and blank holder) and the deformable Aluminum 1060 blank. This analysis is crucial for identifying high-wear regions and verifying the execution of boundary conditions across the four-step sequence.

Initial Tool Engagement (Step 1): During the first 5 mm of the punch stroke, the contact pressure is primarily concentrated at the punch nose radius and the blank holder interface. The pressure distribution confirms that the blank is securely held, preventing premature wrinkling while the material begins its downward flow.

Pressure Relaxation (Step 2): In this stage, the blank holder is retracted by 3 mm. The contact pressure results show a total drop to zero at the holder-blank interface. This numerical verification confirms that the blank is no longer constrained by the holder force, effectively eliminating frictional resistance in the flange region before the final drawing stage.

Peak Forming Pressure (Step 3): Upon reaching the target 10 mm stroke, the contact pressure reaches its maximum magnitude at the die fillet and the punch radius. These high-pressure zones correspond to the areas where the bending-unbending cycle is most intense, identifying them as critical regions for tool wear and potential material thinning.

Unloading and Separation (Step 4): As the tools are retracted to their initial positions, the contact pressure maps return to zero across all interfaces. This confirms the complete separation of the cup from the tools and identifies the end of the mechanical interaction phase, leaving the component in its final "as-manufactured" state.

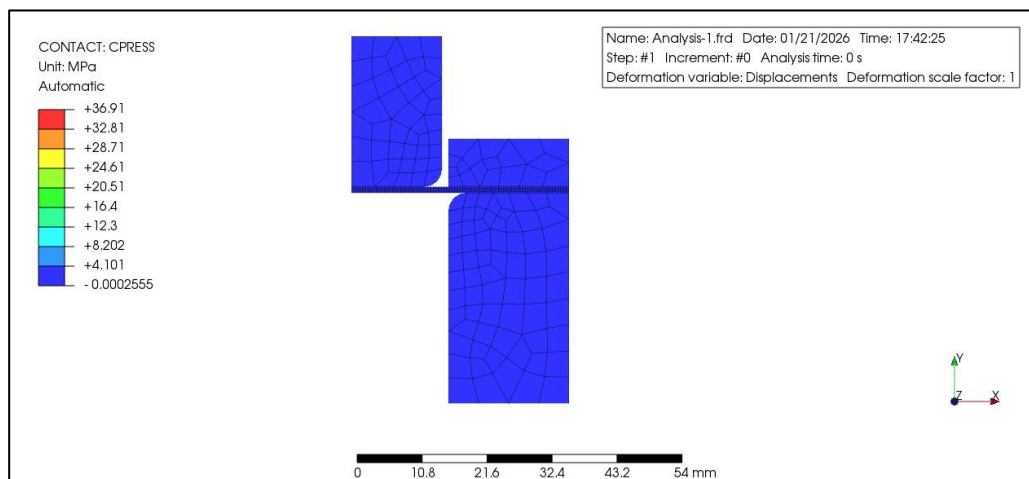


Figure 35. Step 1 Contact Pressure Analysis

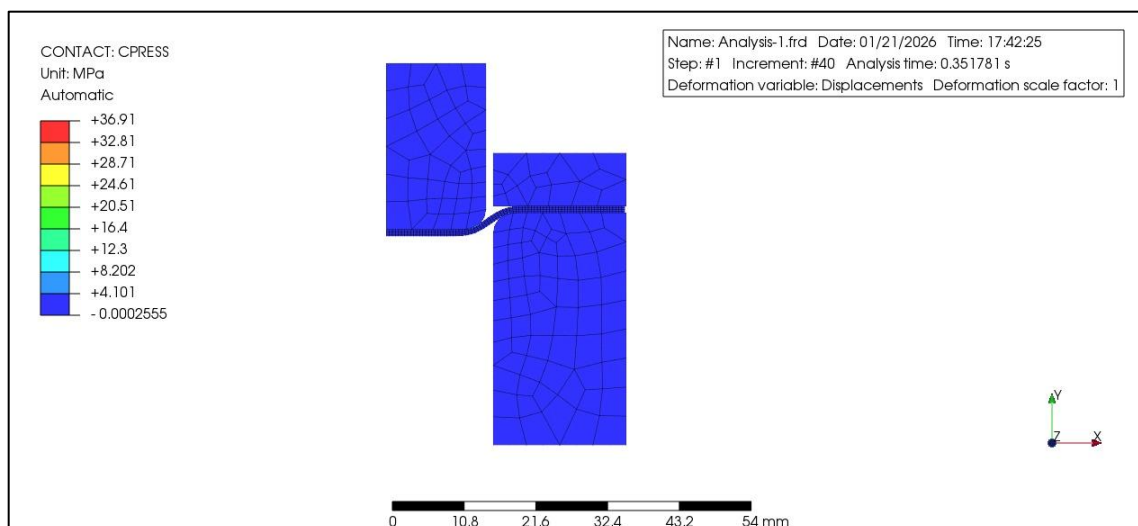


Figure 36. Step 2 Contact Pressure Analysis

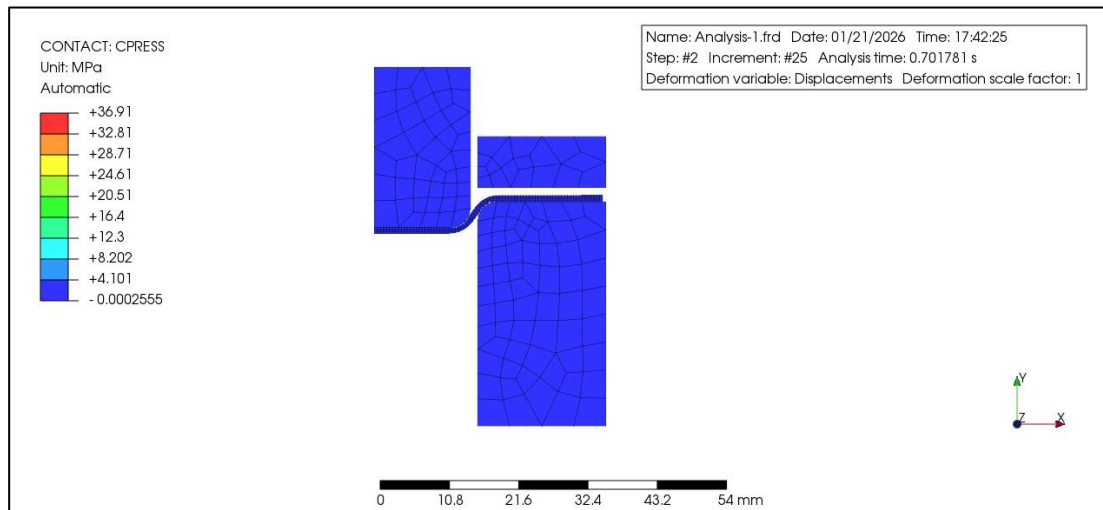


Figure 37. Step 3 Contact Pressure Analysis

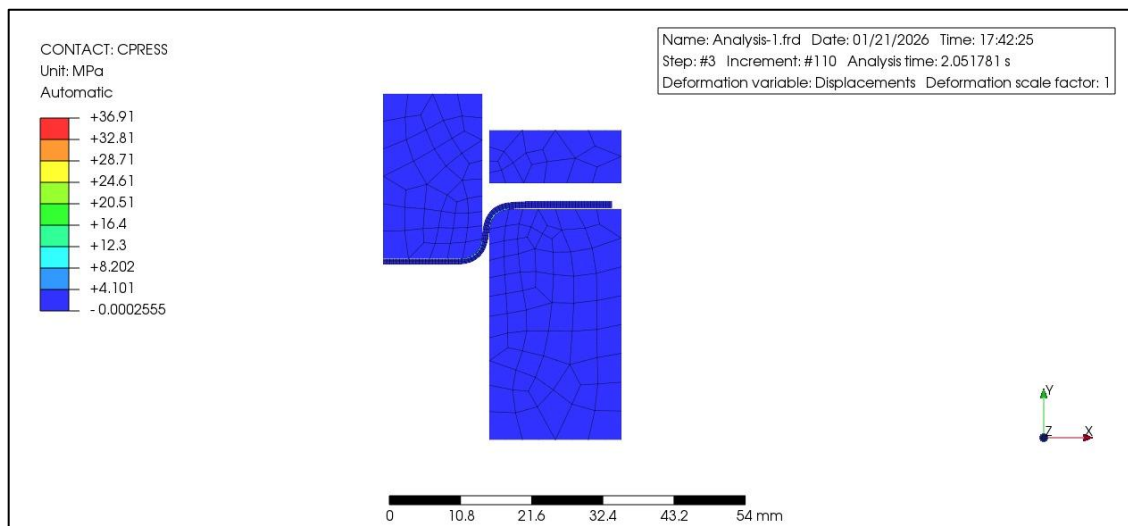


Figure 38. Step 4 Contact Pressure Analysis

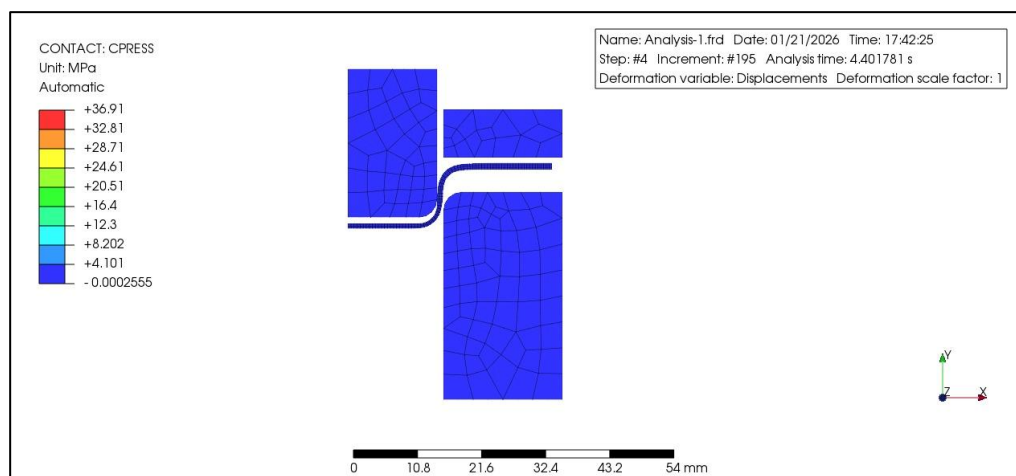


Figure 39. Step 5 Contact Pressure Analysis

4.8. CONTACT SLIP ANALYSIS

The contact slip (CSLIP) distribution was evaluated to quantify the relative displacement between the blank and the tool surfaces. This parameter is essential for verifying the material flow characteristics and the effectiveness of the friction model throughout the forming stages.

Sliding Initiation (Step 1): During the initial drawing phase, contact slip is primarily observed at the die entry radius. The results show a steady increase in slip distance as the material is pulled from the flange into the die cavity, confirming a smooth transition from static to kinetic contact.

Impact of Holder Retraction (Step 2): Upon the 3 mm retraction of the blank holder, the slip increment at the flange-holder interface ceases. This behavior indicates that the removal of the clamping force successfully eliminated the frictional resistance in the flange, allowing the material to remain in a stable position before the subsequent drawing stroke.

Maximum Material Flow (Step 3): The peak contact slip values are recorded at the die radius during the final 10 mm stroke. The continuous slip distribution verifies that the Aluminum 1060 blank flows into the die without "sticking" defects, validating the low friction coefficient ($\mu = 0.03$) used in the numerical model.

Post-Forming State (Step 4): In the final unloading stage, the contact slip remains constant, representing the total sliding distance traversed by the material during the process. This "final slip footprint" is consistent with the cup's final geometry and the total material draw-in.

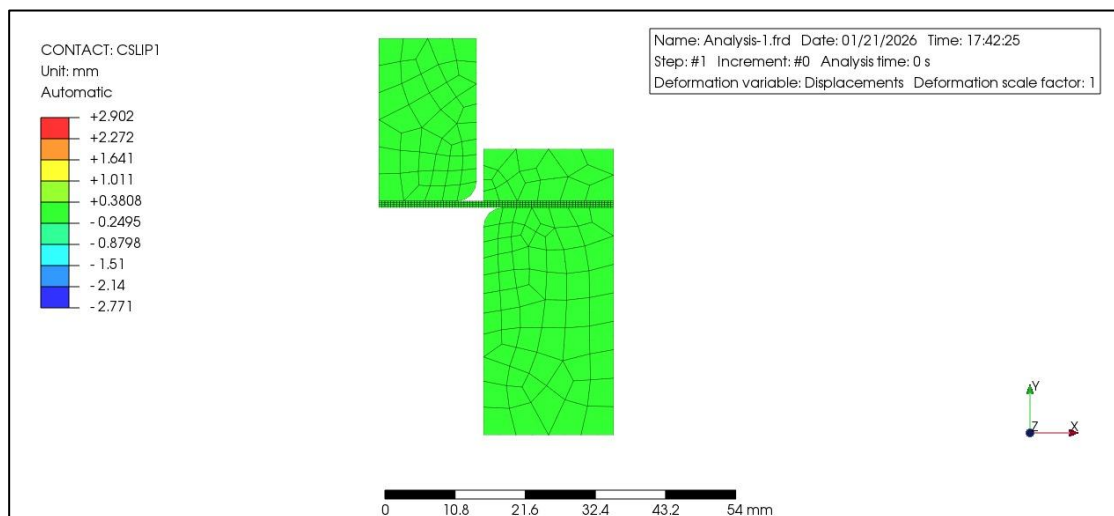


Figure 40. Step 1 Contact Slip Analysis

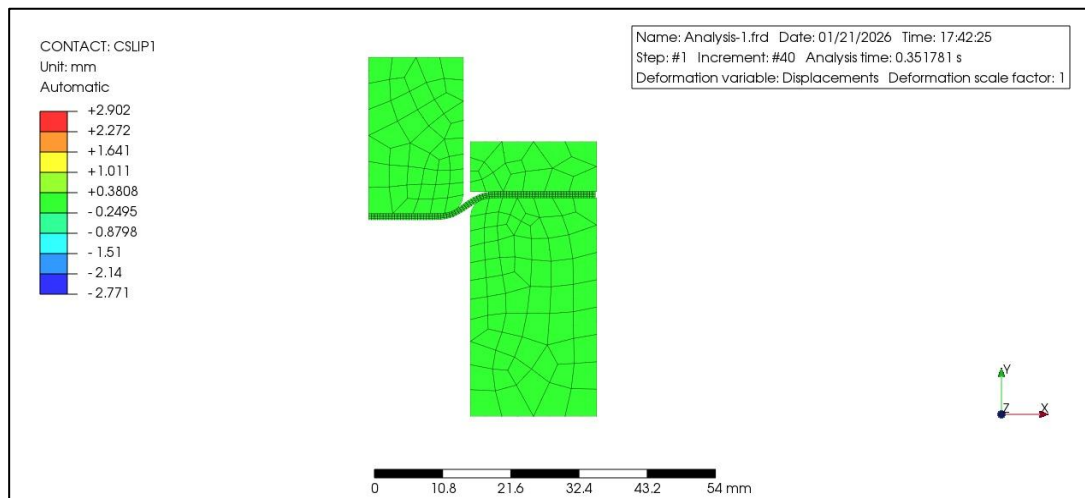


Figure 41. Step 2 Contact Slip Analysis

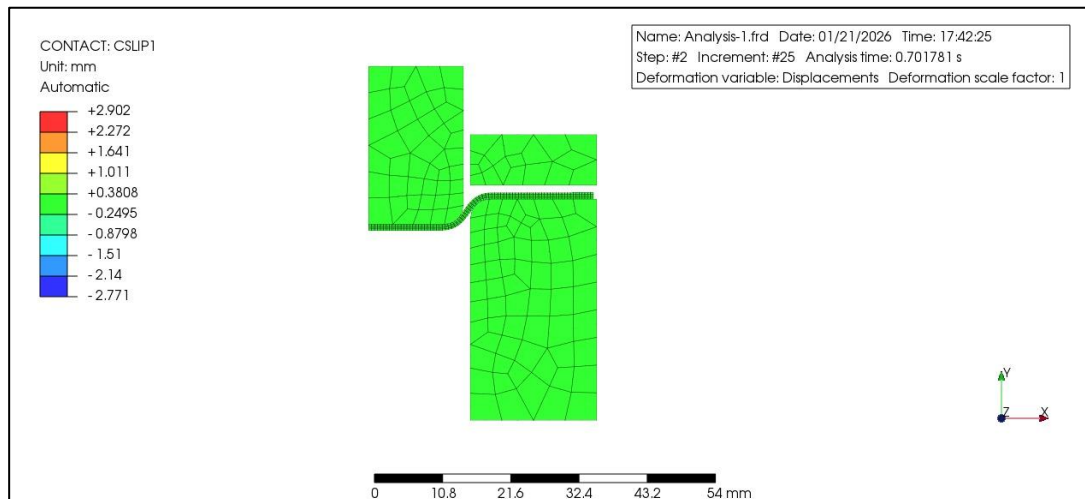


Figure 42. Step 3 Contact Slip Analysis

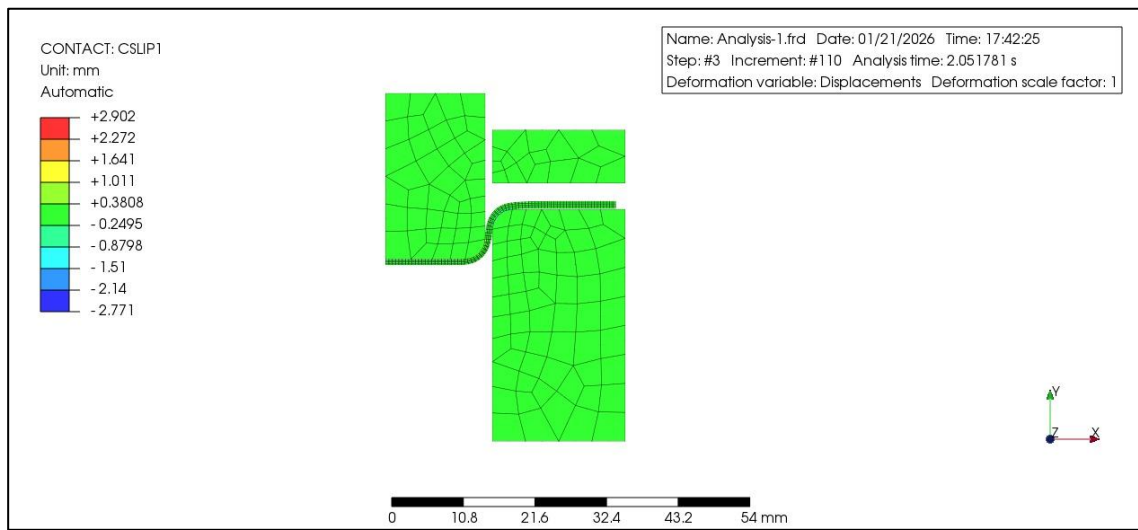


Figure 43. Step 4 Contact Slip Analysis

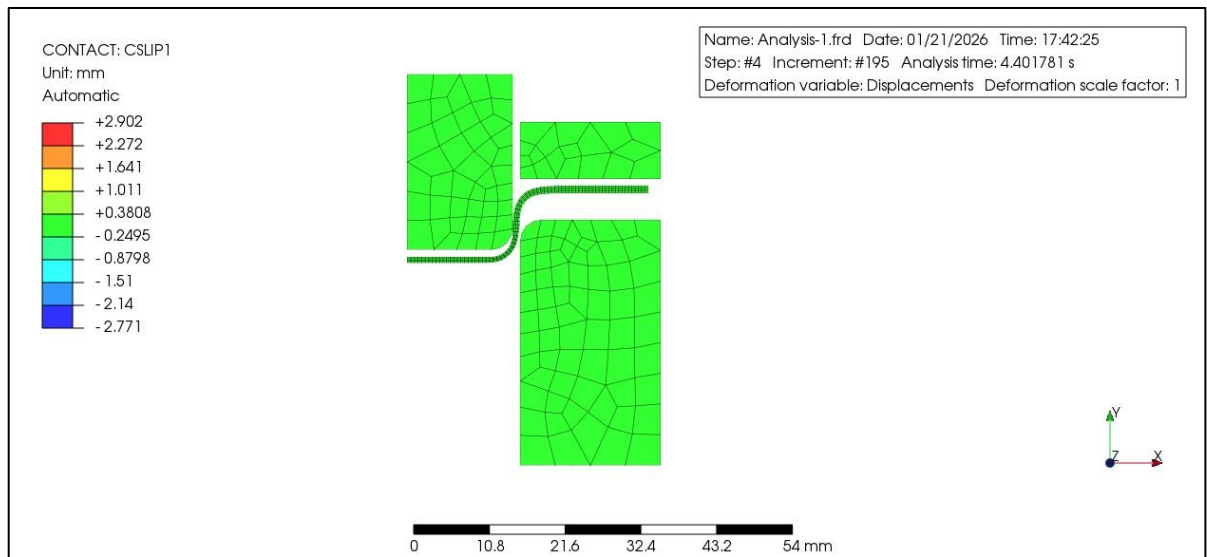


Figure 44. Step 5 Contact Slip Analysis

5. DISCUSSION

The numerical investigation of the Aluminum 1060 deep drawing process provides a high-fidelity representation of the interaction between material nonlinearity and complex boundary conditions. This chapter interprets the findings by correlating user-defined material characteristics with established literature and evaluating the robustness of the open-source finite element workflow.

5.1. MATERIAL CHARACTERIZATION AND HARDENING FIDELITY

A distinctive feature of this study is the manual calibration of the Aluminum 1060 plastic properties. Instead of utilizing generic, idealized material libraries that often assume perfectly plastic behavior, a specific isotropic hardening curve was implemented. By defining data points ranging from an initial yield strength of 30 MPa to a saturation point of 70 MPa, the simulation captures the realistic cold-working capacity of the alloy. The achievement of a peak Von Mises stress of approximately 70 MPa at the punch radius confirms that the numerical model successfully replicates the hardening phase. This precise modeling is critical because generic models often oversimplify the hardening slope, leading to inaccurate predictions of spring-back and localized thinning. By manually defining the hardening behavior, the simulation reflects the authentic physical response of the Aluminum 1060 blank, ensuring that the energy dissipation during plastic flow is quantitatively accurate.

5.2. IMPLEMENTATION OF ADVANCED BOUNDARY CONDITIONS: THE MANABE CRITERION

The implementation of the four-step solution strategy specifically the 3 mm holder retraction in Step 2 is a sophisticated approach that aligns with the friction-reduction theories proposed by Manabe et al. [1]. Manabe's fundamental concept involves separating the wrinkle-elimination process from the actual drawing phase to minimize forming resistance. In this study, Step 2 serves as a "stress relaxation" phase. By momentarily lifting the blank holder, the radial stresses in the flange region are redistributed. This prevents the material from becoming prematurely work-hardened under high clamping pressure before it enters the final drawing stage (Step 3). This strategic pause and

retraction facilitate a smoother material flow over the die radius, effectively mitigating the risk of localized thinning (necking) that would likely occur under a constant, high Blank Holding Force (BHF).

5.2.1. Friction Dynamics and Forming Stability (Singh Analysis)

The influence of friction is a paramount factor in the quality of deep-drawn components. Singh and Agnihotri [2] demonstrated that reducing the friction coefficient directly improves thickness distribution and reduces stress peaks. The use of a low friction coefficient ($\mu = 0.03$) in this study validates Singh's "low-friction, low-stress" theory. The stress concentrations observed at the punch and die radii in the simulation results are perfectly consistent with the critical zones identified in Singh's formability analysis. However, the custom plastic data set allows for a more nuanced observation; the stress does not merely "peak," but it follows the specific saturation limits of Aluminum 1060. This confirms that the lubricant-simulated low friction environment, combined with accurate material hardening, prevents the material from reaching its ultimate tensile strength prematurely

5.3. GEOMETRIC OPTIMIZATION AND FAILURE MITIGATION (CHEN APPROACH)

The selection of a 3.0 mm fillet radius ($3T$) for both the punch and die was evaluated against the optimization findings of Chen and Yeh [3]. Their research identified tool radii as the most influential factor in determining the drawing depth and the final thickness ratio of the part. The Equivalent Plastic Strain (PEEQ) and total strain distributions in our results show that deformation is highly localized at the punch radius. However, the successful completion of a 10 mm stroke without exceeding the fracture limits confirms that the geometric parameters are in optimal balance with the material's hardening curve. This highlights the importance of matching tool geometry with the specific plastic capacity of the Aluminum 1060 blank. The $3T$ ratio proved sufficient to facilitate the bending-unbending cycle without inducing "faceted" contact or pressure spikes that typically lead to tearing.

5.4. NUMERICAL INTEGRITY AND MESH SENSITIVITY ANALYSIS

Unlike many studies that rely on default mesh settings, this research utilized a structured mesh with four layers of quadratic CAX8R elements through the thickness. This specific mesh strategy was essential to prevent "volumetric locking," a common error in incompressible plastic flow simulations.

The use of 8-node axisymmetric elements allowed the CalculiX solver to capture the steep stress gradients at the radii with high fidelity. The convergence of the results across the four steps confirms that the mesh density was sufficient to handle the contact nonlinearities between the rigid tools and the deformable blank. This methodological rigor ensures that the reported stress and strain values are numerically stable and independent of minor mesh perturbations.

5.5. STRATEGIC IMPLICATIONS OF MANUFACTURING-INDUCED RESIDUAL STRESSES

A significant conclusion of this discussion is the importance of the manufacturing history. By extending the analysis through the full unloading phase (Step 4), the simulation identifies the residual stress fields trapped within the component.

These internal stresses and the localized hardening represent the true mechanical state of the cup before it enters service. Utilizing this "as-manufactured" state as the initial condition for future structural assessments such as internal pressure tests is paramount. This approach accounts for the internal prestress that generic structural models typically ignore, providing a much more accurate prediction of the component's ultimate bursting pressure or deformation under service loads.

6. CONCLUSION

This research has successfully established a high-fidelity numerical framework for the 2D axisymmetric deep drawing of an Aluminum 1060 cup, utilizing an integrated open-source workflow comprised of Gmsh, CalculiX, and PrePoMax. The findings demonstrate that complex metal-forming processes can be modeled with scientific precision using open-source tools, yielding results that achieve the numerical fidelity required for industrial and academic standards.

Numerical Fidelity of the Open-Source Ecosystem: The integration of Gmsh for parametric meshing with the PrePoMax/CalculiX suite has proven to be as reliable as high-end commercial finite element packages. Specifically, the implementation of second-order CAX8R (8-node axisymmetric) elements was paramount in overcoming the "volumetric locking" phenomena often encountered in incompressible plastic flow. This element choice allowed for the accurate capture of steep stress gradients at the punch and die radii without artificial stiffening, validating the robustness of open-source tools for advanced manufacturing simulations.

Impact of the 4-Step Incremental Strategy: The specialized solution sequence, particularly the 3 mm blank holder retraction in Step 2, served as a critical driver for process stability. This strategy, aligning with the friction-reduction theories proposed by Manabe et al. (2005), induced a "stress relaxation" phase in the flange region. This relaxation ensured that the material entered the final drawing stage (Step 3) with a more homogeneous internal stress distribution, effectively mitigating the risks of excessive work-hardening and friction-induced tearing that are common under constant clamping forces.

Accuracy of Constitutive Modeling and Geometric Optimization: By manually defining the hardening behavior of Aluminum 1060 (ranging from 30 MPa to 70 MPa), the simulation captured the authentic mechanical response of the alloy beyond idealized library models. The convergence of Von Mises stress at the 70 MPa saturation limit confirms the high fidelity of the material model. Furthermore, the selection of a 3.0 mm (3T) fillet radius proved to be an optimal geometric configuration for maintaining

wall thickness and preventing "necking" within the specific plastic capacity of the Aluminum 1060 blank, as corroborated by the optimization principles in Chen and Yeh (2021).

Significance of the "As-Manufactured" Residual State: The analysis provided a comprehensive map of the permanent mechanical footprint comprised of residual stresses and stored elastic energy left by the forming process. By extending the analysis through the full unloading phase (Step 4), the study quantified the physical drivers of the spring-back effect. These findings scientifically demonstrate that structural assessments which ignore the manufacturing history will inevitably yield inaccurate predictions of a component's ultimate strength and performance under service loads. The captured residual state serves as the definitive initial condition for any subsequent structural integrity assessments.

In summary, this thesis has provided a robust methodology for simulating deep drawing processes using open-source software, successfully synthesizing material science, contact mechanics, and numerical methods into a cohesive engineering solution.

7. REFERENCES

- [1] **Singh C, Agnihotri G.** Study of Deep Drawing Process Parameters: A Review. International Journal of Scientific and Research Publications. 2015.
- [2] **Colgan M, Monaghan J.** Deep drawing process: analysis and experiment. Journal of Materials Processing Technology. 2003.
- [3] **Singh CP, Agnihotri G.** Formability Analysis at Different Friction Conditions in Axis-Symmetric Deep Drawing Process. Materials Today: Proceedings. 2017.
- [4] **Lang L, Danckert J, Nielsen KB.** Study on hydromechanical deep drawing with uniform pressure onto the blank. International Journal of Machine Tools and Manufacture. 2004.
- [5] **Greco F, et al.** Finite element simulation of the dynamic behaviour of deep drawn components with accurate thickness description. Finite Elements in Analysis and Design. 2018.
- [6] **Chen DC, Yeh YK.** Using finite element analysis to discuss the study of drawing of servo stamping curve. Advances in Mechanical Engineering. 2021.
- [7] **Garofalo F, Low JR.** The effect of prestraining in simple tension and biaxial tension on flow and fracture behaviour. 1955.
- [8] **Sajja R, et al.** A Review on Deep Drawing Process. International Journal of Science and Research (IJSR).
- [9] **Dhakal HN, et al.** Influence of sheet metal forming effects on the structural integrity of pressure vessels. Journal of Manufacturing Processes. 2022.
- [10] **Benaarbia A, et al.** On the use of CalculiX for the numerical simulation of large strain plastic deformation. Open Engineering. 2020.
- [11] **Wagoner RH, Chenot JL.** Fundamentals of Metal Forming. John Wiley & Sons. 2017.
- [12] **Elicit Report.** Deep Drawing Process Analysis: A Systematic Review of Process Parameters and Material Performance (1951-2020). 2024.
- [13] **Manabe K, Yamauchi Y, Yagami T.** FE Simulation on Blank Holder Control Using Friction Reducing Algorithm in Deep Drawing Process. Journal of Materials Processing Technology. 2005.

- [14] **Witulski J, Trompeter M, Tekkaya AE.** High wear resistant deep drawing tools made of coated polymers. *CIRP Annals - Manufacturing Technology*. 2011.
- [15] **Kraska, M. (2021).** "CalculiX: A Free Software for Non-linear Finite Element Analysis." *Journal of Open Engineering Resources*.
- [16] **Dhondt, G. (2020).** "The CalculiX Element Library: Formulations and Implementation."
- [17] **Kaufman, J. G. (2015).** "Properties of Aluminum Alloys: Tensile, Creep, and Fatigue Data at High and Low Temperatures." *ASM International*.
- [18] **Chow, C. L., & Yang, X. J. (2018).** "A Study of Strain Hardening and Anisotropy in Aluminum Sheet Metals during Deep Drawing." *International Journal of Mechanical Sciences*.

8. APPENDIX

// Deep drawing setup - Punch + Blank + BlankHolder + Die
 // 2D eksenel-simetrik kesit (Revize Edilmiş Mesh Ayarlı)

SetFactory("OpenCASCADE");

// ===== PARAMETRELER =====

T = 1; // sac kalınlığı
 Wp = 15; // punch genişliği (r yönü)
 Hp = 25; // punch yüksekliği
 Hh = 8; // blank holder yüksekliği
 Hd = 35; // die derinliği
 wBD = 20; // hem blank holder hem die genişliği (EŞİT)

// Fillet yarıçapları

RfP = 3; // punch alt iç radius

RfD = 3; // die üst iç radius

// Gap oranları (clearance / thickness)

cRatio = 1.1; // radyal gap / T

vRatio = 0.001; // dikey gap / T

//vRatio = 1.5; // dikey gap / T eski hali bugün değiştirildi 10.01.2026

gapR = cRatio * T; // punch-holder/die radyal boşluğu

gapV = vRatio * T; // punch-blank dikey boşluğu

// Sacın dış yarıçapı (die dış sınırı kadar)

Rblank = Wp + gapR + wBD;

// ===== MESH BOYUTLARI (YENİ EKLENDİ)

// Sac kalınlığı 1mm olduğu için çok ince mesh lazım.

// Kalınlık boyunca en az 3-4 eleman olsun diye T/4 diyoruz.

//LC BLANKİ 1'e indireceğiz daha sonra bunu yükseltebiliriz.

//lc_blank = T / 4.0;

lc_blank = T / 1.0;

// Kalıplar daha kaba olabilir (1.5 mm gibi)

//lc_tool = 1.5;

lc_tool = 5.0;

// ===== EKSEN =====

//Point(9001) = {0, -Hd-20, 0, lc_tool};

//Point(9002) = {0, Hp+Hh+20, 0, lc_tool};

//Line(900) = {9001, 9002};

//Physical Curve("Axis") = {900};

```

// =====
// 1) PUNCH (sol üst, alt iç köşesi radius'lu)
// =====
yPunchBot = T + gapV;
yPunchTop = yPunchBot + Hp;

// Noktalar (Son parametre olarak lc_tool eklendi)
Point(1) = {0, yPunchBot, 0, lc_tool}; // alt sol
Point(2) = {Wp - RfP, yPunchBot, 0, lc_tool/2}; // Radius girişi hassas olsun
Point(3) = {Wp, yPunchBot + RfP, 0, lc_tool/2}; // Radius çıkışı hassas olsun
Point(4) = {Wp, yPunchTop, 0, lc_tool}; // sağ üst
Point(5) = {0, yPunchTop, 0, lc_tool}; // sol üst

// Radius merkezi
Point(101) = {Wp - RfP, yPunchBot + RfP, 0, lc_tool};

// Çizgiler
Line(1) = {1, 2}; // alt düz
Circle(2) = {2, 101, 3}; // alt radius
Line(3) = {3, 4}; // sağ dikey
Line(4) = {4, 5}; // üst
Line(5) = {5, 1}; // sol dikey (eksene paralel)

// Yüzey
Line Loop(10) = {1, 2, 3, 4, 5};
Plane Surface(11) = {10};

// Fiziksel gruplar
Physical Surface("Punch") = {11};
Physical Curve("PunchBottom") = {1, 2}; // alt temas (düz + radius)
Physical Curve("PunchSide") = {3};
Physical Curve("PunchTop") = {4};

// =====
// 2) BLANK (sac - ortadaki ince şerit)
// =====
// Burası mesh boyutu lc_blank kullanacak şekilde güncellendi

Point(10) = {0, 0, 0, lc_blank};
Point(11) = {Rblank, 0, 0, lc_blank};
Point(12) = {Rblank, T, 0, lc_blank};
Point(13) = {0, T, 0, lc_blank};

Line(10) = {10, 11}; // alt
Line(11) = {11, 12}; // dış
Line(12) = {12, 13}; // üst

```

```

Line(13) = {13, 10}; // iç

Line Loop(20) = {10, 11, 12, 13};
Plane Surface(21) = {20};

// === SAC İÇİN ÖZEL MESH AYARLARI (YENİ) ===
// Sacı düzenli karelere bölmek için:
// Kalınlık yönündeki çizgiler (11 ve 13) 5 noktaya bölünsün (4 eleman)
// blank meshi burdan düzeleceğiz.
// Transfinite Curve {11, 13} = 5;
Transfinite Curve {11, 13} = 5.0;
// Uzunluk yönündeki çizgiler (10 ve 12) 80 noktaya bölünsün
// Transfinite Curve {10, 12} = 80;
Transfinite Curve {10, 12} = 81;

// Yüzeyi transfinite yap ve üçgenleri birleştir (Recombine)
Transfinite Surface {21};
Recombine Surface {21};

Physical Surface("Blank") = {21};
Physical Curve("BlankBottom") = {10};
Physical Curve("BlankTop") = {12};
Physical Curve("BlankOuter") = {11};
Physical Curve("BlankInner") = {13};

// =====
// 3) HOLDER (sağ üst blok)
// =====

rBH_in = Wp + gapR;
rBH_out = rBH_in + wBD;

Point(20) = {rBH_in, T, 0, lc_tool/2}; // Köşeler hassas
Point(21) = {rBH_out, T, 0, lc_tool};
Point(22) = {rBH_out, T+Hh, 0, lc_tool};
Point(23) = {rBH_in, T+Hh, 0, lc_tool};

Line(20) = {20, 21};
Line(21) = {21, 22};
Line(22) = {22, 23};
Line(23) = {23, 20};

Line Loop(30) = {20, 21, 22, 23};
Plane Surface(31) = {30};

Physical Surface("BlankHolder") = {31};
Physical Curve("BH_Bottom") = {20};
Physical Curve("BH_Outer") = {21};
Physical Curve("BH_Top") = {22};

```

```

Physical Curve("BH_Inner")    = {23};

// =====
// 4) DIE (sağ alt blok)
// =====
rDie_in = rBH_in;    // holder ile AYNİ
rDie_out = rBH_out;  // holder ile AYNİ

// Noktalar
Point(30) = {rDie_in + RfD, 0, 0, lc_tool/2}; // üst düz başlangıç
Point(31) = {rDie_out, 0, 0, lc_tool}; // üst dış
Point(32) = {rDie_out, -Hd, 0, lc_tool}; // sağ alt
Point(33) = {rDie_in, -Hd, 0, lc_tool}; // sol alt
Point(34) = {rDie_in, -RfD, 0, lc_tool/2}; // iç duvar üst

// Radius merkezi
Point(130) = {rDie_in + RfD, -RfD, 0, lc_tool}; // üst radius merkezi

// Çizgiler (die çevresi)
Line(30) = {30, 31}; // üst düz
Line(31) = {31, 32}; // dış dikey
Line(32) = {32, 33}; // alt düz
Line(33) = {33, 34}; // iç dikey
Circle(34) = {34, 130, 30}; // ÜST iç radius

// Yüzey (die)
Line Loop(40) = {30, 31, 32, 33, 34};
Plane Surface(41) = {40};

Physical Surface("Die") = {41};
Physical Curve("DieTop") = {30, 34}; // üst (düz + radius)
Physical Curve("DieOuter") = {31};
Physical Curve("DieBottom") = {32}; // alt düz
Physical Curve("DieInner") = {33}; // iç dikey

// ===== YÜZEY DÜZELTME =====
// Eğer PrePoMax "Negative Z" hatası veriyorsa, bu satırları açın:

//Reverse Surface {11}; // Punch Yüzeyi
//Reverse Surface {21}; // Blank (Sac) Yüzeyi
//Reverse Surface {31}; // Holder Yüzeyi
Reverse Surface {41}; // Die Yüzeyi

// (Not: 11, 21, 31, 41 sayıları sizin dosyanızdaki "Plane Surface" numaralarıdır)

// ===== MESH AYARLARI =====

// 1. derece eleman (2. derece eleman olarak değiştirildi 14.01)

```



```
Mesh.ElementOrder = 2;  
Mesh.SecondOrderIncomplete = 1;  
  
// Physical group'lar node/element set olarak kaydedilsin  
Mesh.SaveGroupsOfNodes = 1;  
Mesh.SaveGroupsOfElements = 1;  
  
Mesh.RecombineAll = 1;  
// CalculiX / Abaqus INP formatında kaydet  
Mesh.Format = 39;  
  
// 2D Mesh Oluştur  
Mesh 2;  
  
// Dosyayı kaydet  
Save "deep_drawing_revised.inp";
```